pyAmpliCol 0.1.0

Methodology and Performance Report

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Release 0.1.0

Measurement profile: `x86_EPYC`

Abstract

This report presents the physics coverage, numerical validation, and performance of `pyAmpliCol` 0.1.0. We compare compiled, eager, and recurrence evaluation for the built-in Standard Model and for the same model imported from UFO or serialized JSON, together with scalar-contact and scalar-gravity benchmarks. Every reported timing is associated with a documented software revision, hardware profile, numerical configuration, and validation reference appropriate to the corresponding data set.

Contents

1 Scope

`pyAmpliCol` generates and evaluates tree-level scattering amplitudes. Compiled mode specializes a shared-current directed acyclic graph (DAG) to a process. Eager mode evaluates the same graph with reusable model kernels. Recurrence mode constructs a compact current schedule directly. The three modes implement the same physics normalization, colour treatment, helicity sums, and resolved observables. The original `AmpliCol` program and its use in event generation are described in Ref. [5].

The report has three purposes:

1. define the observables and conditions required for a reproducible benchmark;
2. preserve a stable matrix of representative process families and colour approximations; and
3. present performance only after numerical agreement has been verified.

In each architecture-specific report, an N/A entry denotes a process/multiplicity combination outside the declared scope or an observable that does not apply. Every applicable cell across the complete declared multiplicity range must instead contain a validated measurement or an documented resource-limit status. A measurement that does not pass numerical validation is labelled explicitly and never replaced by a timing value. No value is copied, interpolated, or estimated from a neighbouring cell.

2 Current Recursion and Model Lowering

2.1 Shared-current DAG

For an external subset I, particle state t, and local spin or flow label χ, an off-shell current has the schematic recursion

$$J^{\alpha}_{t,\chi}(I) = P^{\alpha}{}_{\beta}(t, p_I) \sum_{I=A\dot{\cup}B} V^{\beta}{}_{\gamma\delta}(t_A, t_B \to t) J^{\gamma}_{t_A,\chi_A}(A) J^{\delta}_{t_B,\chi_B}(B), \qquad p_I = \sum_{i\in I} p_i. \tag{1}$$

Current identity includes the particle, external labels, helicity ancestry, chirality or spin state, flavour and charge flow, colour state, momentum mask, and any auxiliary-current kind. Contributions with the same complete identity share one node. A topological sweep therefore evaluates each retained current once per phase-space point and reuses it in all dependent amplitudes. This is a Berends–Giele-type construction [1].

The Universal Feynman Output format (UFO; originally Universal FeynRules Output) describes particles, parameters, interactions, Lorentz tensors, colour tensors, and propagators in a model-independent form [3, 4]. `pyAmpliCol` normalizes a trusted UFO Python module, or its own serialized JSON representation of that model, to the same canonical representation used by process generation. The UFO module is executable Python; the corresponding serialized JSON form is data-only.

Optimizations of imported models are based on exact colour, Lorentz, propagator, and coupling identities rather than particle names or numerical codes. For Standard-Model validation processes, the built-in and imported routes must produce equivalent current, interaction, amplitude, and colour content. Algebraic reuse is allowed only when the relevant identity has been established exactly.

2.2 Contact terms and spin two

A supported vertex with valence above three is lowered to a deterministic tree of non-propagating auxiliary currents. The construction inserts the original coupling once and introduces no pole. Direct contraction of the original contact tensor is the validation reference for the scalar-contact ladder.

The scalar-gravity ladder exercises rank-two external states and momentum-dependent interactions. For a massless graviton, the helicity-two polarization tensor is represented as

$$\epsilon^{\mu\nu}_{\pm 2} = \epsilon^{\mu}_{\pm}\epsilon^{\nu}_{\pm}. \tag{2}$$

Validation checks symmetry, transversality, tracelessness, normalization, and agreement between ordinary and higher-precision evaluation.

3 Colour and Runtime Semantics

The process matrices use three colour contracts:

LC Leading-colour ordered amplitudes with physical colour-flow selectors [2].
NLC Every colour-metric entry whose leading power is at most two below the leading power of the complete basis.
Full The complete supported sparse colour metric for the generated amplitude basis.

For NLC, "retained" applies to a colour-matrix *entry*, not to a truncated numerical coefficient. Write the exact overlap of two colour-flow tensors as

$$C_{ab}(N_c) = \sum_p c_p^{(ab)} N_c^p, \qquad p_{ab} = \max\{p : c_p^{(ab)} \neq 0\}, \tag{3}$$

and let P be the leading power of the complete colour basis. The sparse NLC metric is

$$C_{ab}^{\mathrm{NLC}} = \begin{cases} C_{ab}(3), & p_{ab} \geq P - 2, \\ 0, & p_{ab} < P - 2. \end{cases} \tag{4}$$

Thus every qualifying entry keeps all of its finite-N_c terms. For three open fundamental lines and one adjoint, $P = 4$. Representative retained off-diagonal overlaps are

exact overlap	**highest power**	**NLC at $N_c = 3$**	**full at $N_c = 3$**
$-N_c^3 + N_c$	3	-24	-24
$N_c^2 - 1$	2	8	8

The powers and coefficients are retained exactly. When only the upper-triangular part of the colour metric is stored, off-diagonal interference terms receive the usual factor of two; this factor is not part of the colour coefficient in Eq. (4).

The summed observable returns one matrix element per point. Leading-colour resolved values have axes `(point, helicity, flow)`; NLC and full-colour values have one contracted colour component per helicity. Summing all non-point axes must reproduce the directly evaluated total. Selectors refer to physical helicities and flows.

All `pyAmpliCol` binary64 interfaces use the same native numerical core. Cross-interface agreement therefore checks data conventions, while a separately generated value is required for an independent physics comparison.

The JIT backend stores a portable SymJIT application for binary64 evaluation; the receiving host creates code appropriate for its processor. ASM and C++ payloads are target-specific. Evaluation above binary64 precision uses the retained symbolic evaluator state.

4 Benchmark Methodology

4.1 Configuration contract

Each measurement begins from an immutable resolved configuration. Unless a table states otherwise, measurements use a batch size of 128, two warmup runs, at least five timed samples, a five-second sampling target, and one CPU core. Compiled process matrices use JIT O3; eager and recurrence matrices use portable JIT O2 prepared models. Explicitly labelled rows in the dedicated Z ladder use JIT O1, JIT O3, ASM O3, or C++ O3. Backend-specific optimization settings are retained with the measurement record.

Generation time begins with the resolved model input required by an execution mode and ends with a validated, loadable process representation. Compiled mode includes process-specific evaluator construction and backend compilation. Eager and recurrence modes include construction and serialization of their respective numerical schedules. Preparation of a reusable built-in or UFO/JSON model bundle occurs before this timer and is therefore excluded from the process-generation values in the tables.

Runtime measurements use the summed matrix element. The reported native wall time starts after caller-language array conversion and includes source construction, numerical evaluation, selector handling, and final reduction. Caller-language packing is excluded. When a backend exposes a separately measured execution attribution, the tables show it alongside wall time; otherwise the field is marked NOT EXPOSED and is never inferred or replaced by zero. Comparisons use the same process, colour accuracy, parameters, phase-space point, precision, selectors, and normalization.

4.2 Benchmark platform

This document is an architecture-specific edition. The recorded hardware, operating system, and processor summary is Linux 6.12.63; x86_64; AMD EPYC 9754 128-Core Processor. The measurement profile is `x86_EPYC`; its recorded software environment is CPython 3.11.15; pyAmpliCol 0.1.0.dev0+candidate.d2678d7f1377; NumPy 2.4.2; native x86_64-unknown-linux-gnu. Measurements from different processor architectures are reported separately and are never combined in a summary ratio. Unless a table states otherwise, generation and evaluation use one CPU core and a five-second sampling target.

4.3 Numerical validation

For each Standard-Model process, all compared execution modes and the original `AmpliCol` reference are evaluated at the same deterministic phase-space point with identical numerical parameters. Cross-mode comparisons use relative tolerance 10^{-12} and absolute tolerance 10^{-15}; comparisons with the independent `AmpliCol` result use relative tolerance 10^{-8}. Scalar-model ladders instead compare binary64 evaluation with higher precision and, where applicable, direct contact-tensor contraction. Every result must reproduce its resolved sum and preserve expected structural zeros.

Timing and numerical validation are separate requirements: a timing is reported only when the corresponding matrix element passes all validation checks, and numerical agreement alone does not substitute for a measured generation or runtime value.

5 Standard-Model Process Matrices

The process matrices use recurrence as the primary `pyAmpliCol` execution mode. Built-in-SM and UFO-SM recurrence JIT O2 are compared with independent original-`AmpliCol` results. Separate built-in-SM matrices compare compiled JIT O3 and eager JIT O2 with recurrence JIT O2 for the same process, colour treatment, and observable.

The canonical matrix spans electroweak vector production, charged-current and leptonic channels, heavy-quark production, pure gluons, multi-boson final states, Higgs-associated final states, and three- and four-quark-line cases. Here n is the number of final-state particles. LC coverage retains the declared grid through $n = 9$, with the bounded families ending at the multiplicity stated by the process definition. NLC and full-colour grids retain $n = 1, \dots, 5$. A process/multiplicity combination outside its family definition is marked NOT APPLICABLE; this is table structure, not an unfilled measurement. A complete edition accounts for every applicable cell across the full declared range. Resource-limited cells carry a documented status; every successful cell is numerical and validated. An in-scope result that does not pass numerical validation is identified by a status and does not carry a timing ratio.

The UFO-SM recurrence matrices use the same process definitions, colour contracts, numerical parameters, selectors, and validation points as the built-in-SM recurrence matrices. Their source is the packaged Standard Model loaded through the serialized-JSON interface. This comparison verifies that the imported and built-in model routes implement the same requested physics. Compiled and eager process matrices are built-in-only, isolating differences between execution modes from differences in model loading.

Numerical validation summary

Only measurements whose numerical checks pass are counted below. The scope comprises every applicable report cell across the complete declared multiplicity range.

final-state multiplicity	required	verified	coverage
1	64	28	incomplete (0 censored, 36 open)
2	186	100	incomplete (0 censored, 86 open)
3	206	112	incomplete (0 censored, 94 open)
4	286	142	incomplete (0 censored, 144 open)
5	285	142	incomplete (0 censored, 143 open)
6	165	70	incomplete (0 censored, 95 open)
7	165	73	incomplete (0 censored, 92 open)
8	165	31	incomplete (0 censored, 134 open)
9	124	25	incomplete (0 censored, 99 open)
total	1646	723	incomplete (0 censored, 923 open)

Full-catalog display accounting through $n = 9$: 1646 required measured cells; 316 matrix process/multiplicity positions marked NOT APPLICABLE; and 36 reference execution fields marked NOT EXPOSED. The latter two categories are intentional table structure, not unfilled measurements.

validation comparison	passed	largest relative difference
independent original-`AmpliCol` reference records	235	reference
`pyAmpliCol` versus independent reference	486	2.723×10^{-1}
compiled/eager versus recurrence	2	2.547×10^{-15}
optimized result versus resolved sum	488	7.206×10^{-14}
scalar result versus higher precision	0	—

Profile: `x86_EPYC`. Measured source: source identity is incomplete or mixed. Independent-reference comparisons use a relative tolerance of 10^{-8}; cross-mode, resolved-sum, and higher-precision comparisons use 10^{-12}.

Workloads and denominators

For a reported timing quantity T, each multiplier is

$$r_{\text{candidate/baseline}} = \frac{T_{\text{candidate}}}{T_{\text{baseline}}}. \tag{5}$$

Green, orange, and red denote respectively $r < 1$, $1 \leq r < 2$, and $r \geq 2$. Green therefore means that the candidate is faster than the baseline named in the table heading. The recurrence matrices use original `AmpliCol` as the baseline. The compiled and eager matrices use built-in-SM recurrence JIT O2 as the baseline.

Every LC comparison contains two complete-coverage layouts:

- topology replay, measured with one physical colour flow selected and all retained helicities summed; and
- the all-flow union, measured with one physical helicity selected and all retained colour flows summed.

Neither selection is fixed during generation. Each LC cell reports the two baseline generation quantities and wall times, followed by candidate generation entries and runtime ratios. A separately measured execution attribution is displayed after the wall ratio when both sides expose one; otherwise it is marked NOT EXPOSED. Compiled and eager generation are compared with recurrence using the same layout.

The two original-`AmpliCol` reference workloads have different setup boundaries. For the selected-flow workload, the reported generation quantity is creation of the process library. For the all-flow workload, it is preparation of direct all-flow evaluation. Runtime ratios compare the same physical observable. The all-flow setup quantities are not like-for-like, so the table shows the candidate process-generation time as an absolute value marked `n.c.`; `n.c.` means “not comparable”, so no ratio is formed.

NLC and full colour use one helicity-summed, colour-contracted calculation per mode. Their cells report the absolute baseline generation and wall time, followed by the candidate generation and runtime ratios. No per-flow scalar is defined for these contracted colour accuracies.

The wall quantity is the repeated native evaluation time after caller-language input conversion and is the primary runtime observable. For compiled and eager rows it is the authenticated warmed total-evaluator boundary. A separately measured, narrower execution attribution is supplementary. Its ratio is shown only when candidate and baseline expose compatible attribution boundaries; otherwise it is marked NOT EXPOSED; this label does not hide or invalidate the reported total-evaluator wall time.

Preparation of reusable model-level kernels is excluded from process-generation time. Compiled generation still includes process-specific evaluator construction and backend compilation.

A timing is shown only after recurrence agrees with original `AmpliCol` at the independent-reference tolerance and compiled and eager results agree with recurrence at the stricter cross-mode tolerance. Built-in and UFO-SM recurrence results are also compared directly at that stricter tolerance, rather than relying on their separate agreement with `AmpliCol`. Each result must reproduce its resolved sum. For LC, the common flow–helicity component is compared directly between topology replay and all-flow union, and every `pyAmpliCol` all-flow result is also compared with the matching independently generated `AmpliCol` component. A failed validation has no timing ratio.

Cell and summary layout

contract	baseline entry	candidate entries
LC, upper	topology-replay/all-flow generation quantities in seconds	layout-matched ratios, except for the absolute `n.c.` all-flow candidate against original `AmpliCol`
LC, lower	selected-flow/all-flow wall times in microseconds per point	wall ratio and, when exposed, a separate execution attribution
NLC/full, upper	contracted generation time in seconds	candidate generation ratio
NLC/full, lower	contracted wall time in microseconds per point	wall ratio and, when exposed, a separate execution attribution

Summary rows report the baseline mean over valid candidate–baseline pairs and the ratio of sums $\sum_i T_{\text{candidate},i}/\sum_i T_{\text{baseline},i}$. LC summaries remain separate for topology replay and all-flow union. Structurally inapplicable, resource-limited, unsupported, and validation-failed cells do not contribute to a ratio-of-sums summary.

The first three matrices provide a built-in-SM overview against original `AmpliCol`. For each process, multiplicity, colour treatment, and LC workload, they select the validated pyAmpliCol mode with the smallest wall time. The generation multiplier is followed by `(A)`, `(B)`, or `(C)` for recurrence JIT O2, compiled JIT O3, or eager-DAG JIT O2, respectively. Selection is independent in every cell: the overview records the measured winner rather than prescribing one mode for all processes. For the LC all-flow workload, the displayed setup multiplier spans the different setup boundaries described above and is therefore descriptive, not a like-for-like compiler benchmark.

5.1 Built-in SM best measured mode versus AmpliCol: LC

Built-in SM best measured mode versus AmpliCol LC (block 1 of 3)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	31.9 / 0.000361 (x0.204) (A) (x1.8e+04) (A) 0.0565 / 0.125 (x4.34) (x1.08)	25.6 / 0.000914 (x0.261) (A) (x7.23e+03) (A) 0.366 / 0.288 (x1.87) (x0.969)	27.4 / 0.00149 (x0.246) (A) (x4.45e+03) (A) 1.23 / 0.834 (x1.2) (x0.846)
2 $u\bar{d} \to W^+ + (n-1)g$	31.6 / 0.000382 (x0.209) (A) (x1.71e+04) (A) 0.0321 / 0.124 (x6.02) (x1.1)	26.2 / 0.000411 (x0.252) (A) (x1.76e+04) (A) 0.22 / 0.288 (x2) (x0.979)	27 / 0.000656 (x0.244) (A) (x1e+04) (A) 0.702 / 0.749 (x1.45) (x0.889)
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	26.7 / 0.000927 (x0.244) (A) (x7.21e+03) (A) 0.182 / 0.262 (x1.7) (x0.861)	27.8 / 0.000602 (x0.24) (A) (x1.11e+04) (A) 0.445 / 0.434 (x1.44) (x0.86)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	29.1 / 0.000701 (x0.232) (A) (x1.04e+04) (A) 0.0673 / 0.206 (x2.45) (x1.72)	27.1 / 0.00133 (x0.242) (A) (x4.93e+03) (A) 0.139 / 0.361 (x2.2) (x0.917)
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	29.2 / 0.000861 (x0.228) (A) (x7.67e+03) (A) 0.47 / 0.298 (x1.83) (x0.951)	28.1 / 0.0014 (x0.24) (A) (x4.79e+03) (A) 2.09 / 0.608 (x1.25) (x0.881)
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	27.3 / 0.000944 (x0.256) (A) (x7.62e+03) (A) 0.32 / 0.413 (x2.7) (x0.956)	33.1 / 0.00174 (x0.203) (A) (x3.84e+03) (A) 1.94 / 1.63 (x2.37) (x0.808)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	28.7 / 0.000418 (x0.231) (A) (x1.7e+04) (A) 0.144 / 0.286 (x4.12) (x0.959)	30.7 / 0.000668 (x0.217) (A) (x9.79e+03) (A) 0.63 / 0.945 (x4.82) (x0.859)
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	31.2 / 0.000446 (x0.21) (A) (x1.6e+04) (A) 0.459 / 0.608 (x4.14) (x1.36)	29.7 / 0.00094 (x0.229) (A) (x7.08e+03) (A) 2.4 / 3.77 (x3.37) (x1.32)
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	34.9 / 0.000633 (x0.2) (A) (x1.19e+04) (A) 4.59 / 0.83 (x1.03) (x0.85)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	31.8 (x0.207) / 0.000371 (x1.76e+04) [A:2/B:0/C:0] [A:2/B:0/C:0]	28 (x0.238) / 0.000703 (x9.94e+03) [A:8/B:0/C:0] [A:8/B:0/C:0]	29.5 (x0.227) / 0.00105 (x6.4e+03) [A:9/B:0/C:0] [A:9/B:0/C:0]
summary: wall	0.0443 (x4.95) / 0.124 (x1.09)	0.279 (x2.61) / 0.331 (x1.1)	1.57 (x1.87) / 1.13 (x1.02)

The candidate is selected independently in each cell and workload by the smallest validated wall time. Generation multipliers identify that runtime winner: (A) recurrence JIT O2, (B) compiled JIT O3, and (C) eager-DAG JIT O2. Runtime entries are winner/AmpliCol wall-time multipliers. For the LC all-flow workload, the generation multiplier compares pyAmpliCol process generation with AmpliCol direct-evaluation setup; it is a setup-cost indicator across different boundaries, not a like-for-like compiler benchmark. Summary mode counts use the order A/B/C.

Built-in SM best measured mode versus AmpliCol LC (block 2 of 3)

ID base process	n=4	n=5	n=6
1 $d\bar{d} \to Z + (n-1)g$	64.7 / 0.00227 (x0.154) (A) (x4.12e+03) (A) 4.16 / 2.93 (x1.09) (x0.851)	38.8 / 0.0149 (x0.195) (A) (x612) (A) 12.4 / 15.2 (x0.865) (x0.579)	151 / 0.229 (x0.0803) (A) (x44.8) (A) 157 / 97.2 (x0.227) (x0.754)
2 $u\bar{d} \to W^+ + (n-1)g$	46 / 0.00502 (x0.152) (A) (x1.63e+03) (A) 2.52 / 2.97 (x1.26) (x0.804)	51 / 0.0149 (x0.146) (A) (x599) (A) 8.26 / 15.3 (x1.34) (x0.812)	93.1 / 0.137 (x0.122) (A) (x79.8) (A) 24.2 / 96.3 (x1.85) (x0.826)
3 $d\bar{d} \to e^+e^- + (n-2)g$	76.9 / 0.00193 (x0.101) (A) (x5.78e+03) (A) 1.25 / 0.968 (x1.34) (x0.917)	37.2 / 0.0116 (x0.229) (A) (x727) (A) 3.83 / 3.52 (x1.06) (x0.896)	155 / 0.0965 (x0.0851) (A) (x124) (A) 52.3 / 16.8 (x0.319) (x1.13)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	49.1 / 0.00413 (x0.142) (A) (x2.18e+03) (A) 0.417 / 0.821 (x1.69) (x1.12)	42.5 / 0.0113 (x0.186) (A) (x838) (A) 1.69 / 3.08 (x1.54) (x0.891)	92.1 / 0.0951 (x0.103) (A) (x111) (A) 6.15 / 15.1 (x2.92) (x0.793)
5 $d\bar{d} \to ZZ + (n-2)g$	81 / 0.00209 (x0.112) (A) (x4.36e+03) (A) 6.29 / 1.65 (x1.05) (x0.816)	41.7 / 0.0128 (x0.226) (A) (x713) (A) 16.8 / 6.31 (x0.937) (x0.797)	161 / 0.15 (x0.0757) (A) (x65) (A) 98.3 / 30.7 (x0.442) (x0.596)
6 $gg \to t\bar{t} + (n-2)g$	78 / 0.0032 (x0.106) (A) (x2.88e+03) (A) 6.87 / 8.4 (x5.04) (x1.07)	141 / 0.0261 (x0.077) (A) (x400) (A) 21.1 / 55 (x7.9) (x0.723)	389 / 0.292 (x0.187) (A) (x70.2) (A) 58.4 / 480 (x11.4) (x0.607)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	92.2 / 0.00225 (x0.109) (A) (x4.18e+03) (A) 2.41 / 4.1 (x6.41) (x1.09)	59.5 / 0.0145 (x0.166) (A) (x602) (A) 8.12 / 23.5 (x8.59) (x0.564)	246 / 0.139 (x0.099) (A) (x97.6) (A) 90.8 / 176 (x6.5) (x0.517)
8 $gg \to gg + (n-2)g$	58.9 / 0.00509 (x0.167) (A) (x1.57e+03) (A) 9.23 / 26.1 (x4.18) (x0.877)	56.7 / 0.0716 (x0.272) (A) (x189) (A) 29.4 / 219 (x4.75) (x0.956)	217 / 3.08 (x0.847) (A) (x53.5) (A) 356 / 3.88e+03 (x1.26) (x0.873)
9 $d\bar{d} \to ZZZ + (n-3)g$	37.2 / 0.00348 (x0.227) (A) (x2.08e+03) (A) 12.9 / 1.56 (x0.836) (x0.845)	43.3 / 0.0112 (x0.174) (A) (x684) (A) 33.2 / 3.94 (x0.779) (x0.718)	57.8 / 0.0946 (x0.131) (A) (x82.4) (A) 84.9 / 14.5 (x0.735) (x0.636)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	163 / 0.00196 (x0.0506) (A) (x4.02e+03) (A) 3.78 / 1.79 (x0.853) (x1.26)	103 / 0.011 (x0.0714) (A) (x879) (A) 7.87 / 2.66 (x0.792) (x1.02)	297 / 0.0889 (x0.0304) (A) (x118) (A) 17.1 / 4.91 (x0.668) (x0.858)
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	82.2 / 0.00146 (x0.105) (A) (x6.03e+03) (A) 0.462 / 2.04 (x8.36) (x1.66)	46 / 0.0101 (x0.185) (A) (x884) (A) 1.11 / 9.62 (x38.8) (x0.852)	223 / 0.0996 (x0.0962) (A) (x120) (A) 2.97 / 53.3 (x257) (x1.18)
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	N/A / N/A N/A N/A N/A / N/A N/A N/A
summary: generation	75.4 (x0.114) / 0.00299 (x2.96e+03) [A:11/B:0/C:0] [A:11/B:0/C:0]	60.1 (x0.152) / 0.0191 (x495) [A:11/B:0/C:0] [A:11/B:0/C:0]	189 (x0.181) / 0.409 (x62.8) [A:11/B:0/C:0] [A:11/B:0/C:0]
summary: wall	4.57 (x2.45) / 4.85 (x0.964)	13.1 (x3.45) / 32.4 (x0.863)	86.1 (x2.85) / 442 (x0.832)

The candidate is selected independently in each cell and workload by the smallest validated wall time. Generation multipliers identify that runtime winner: (A) recurrence JIT O2, (B) compiled JIT O3, and (C) eager-DAG JIT O2. Runtime entries are winner/AmpliCol wall-time multipliers. For the LC all-flow workload, the generation multiplier compares pyAmpliCol process generation with AmpliCol direct-evaluation setup; it is a setup-cost indicator across different boundaries, not a like-for-like compiler benchmark. Summary mode counts use the order A/B/C.

Built-in SM best measured mode versus AmpliCol LC (block 3 of 3)

ID base process	n=7	n=8	n=9
1 $d\bar{d} \to Z + (n-1)g$	29.2 / 1.45 (x0.4) (A) (x27.2) (A) 91.9 / 853 (x0.691) (x0.504)	63.5 / N/A (x1.32) (A) N/A 234 / N/A (x0.706) N/A	62.5 / 511 (x18.1) (A) N/A 568 / 2.7e+05 (x0.728) N/A
2 $u\bar{d} \to W^+ + (n-1)g$	46.4 / 1.46 (x0.258) (A) (x31.5) (A) 65.6 / 916 (x0.732) (x0.553)	59.9 / N/A (x1.09) (A) N/A 192 / N/A (x0.618) N/A	45.7 / 511 (x18.1) (A) N/A 435 / 2.19e+05 (x0.678) N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	62.4 / 0.962 (x0.142) (A) (x10.8) (A) 31.7 / 105 (x0.86) (x0.573)	58.8 / N/A (x0.19) (A) N/A 87.6 / N/A (x0.68) N/A	48.5 / N/A (x1.69) (A) N/A 208 / N/A (x0.717) N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	61.5 / 0.932 (x0.158) (A) (x13.2) (A) 19.2 / 96.1 (x0.819) (x0.604)	67.5 / N/A (x0.16) (A) N/A 54.4 / N/A (x1.44) N/A	35.9 / N/A (x1.67) (A) N/A 143 / N/A (x0.721) N/A
5 $d\bar{d} \to ZZ + (n-2)g$	73.7 / 1.02 (x0.144) (A) (x22.7) (A) 111 / 195 (x0.858) (x0.511)	129 / N/A (x0.241) (A) N/A 271 / N/A (x0.901) N/A	141 / N/A (x1.42) (A) N/A 648 / N/A (x0.782) N/A
6 $gg \to t\bar{t} + (n-2)g$	1.25e+03 / 5.84 (x1.65) (A) (x69.9) (A) 154 / 1.23e+04 (x41.4) (x0.624)	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
7 $d\bar{d} \to t\bar{t} + (n-2)g$	136 / 1.58 (x1.2) (A) (x44.3) (A) 65.5 / 2.02e+03 (x19) (x0.511)	491 / 28.7 (x7.74) (A) N/A 170 / 2.41e+04 (x34.5) N/A	4.23e+03 / N/A N/A N/A 582 / N/A N/A N/A
8 $gg \to gg + (n-2)g$	136 / 234 (x32.4) (A) N/A 212 / 7.03e+04 (x7.68) N/A	397 / N/A N/A N/A 539 / N/A N/A N/A	NOT APPLICABLE
9 $d\bar{d} \to ZZZ + (n-3)g$	102 / 0.946 (x0.0866) (A) (x13.1) (A) 203 / 66.6 (x0.75) (x0.472)	268 / 10.9 (x0.0525) (A) (x5.73) (A) 476 / 414 (x0.761) (x0.427)	743 / 124 (x0.0872) (A) (x9.56) (A) 1.25e+03 / 7.52e+03 (x0.727) (x0.264)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	178 / N/A N/A N/A 61.3 / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	154 / 0.872 (x0.0685) (A) (x11.3) (A) 39.3 / 15.4 (x0.581) (x0.717)	193 / N/A (x0.0422) (A) N/A 92.2 / N/A (x0.526) N/A	199 / N/A (x0.0575) (A) N/A 210 / N/A (x1.22) N/A
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	134 / 1.34 (x1.17) (A) (x17.4) (A) 7.91 / 375 (x513) (x0.761)	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
summary: generation	198 (x3.14) / 1.64 (x39.9) [A:11/B:0/C:0] [A:10/B:0/C:0]	166 (x3.02) / 10.9 (x5.73) [A:8/B:0/C:0] [A:1/B:0/C:0]	182 (x1.86) / 124 (x9.56) [A:7/B:0/C:0] [A:1/B:0/C:0]
summary: wall	91 (x13.7) / 1.7e+03 (x0.602)	197 (x4.41) / 414 (x0.427)	494 (x0.76) / 7.52e+03 (x0.264)

The candidate is selected independently in each cell and workload by the smallest validated wall time. Generation multipliers identify that runtime winner: (A) recurrence JIT O2, (B) compiled JIT O3, and (C) eager-DAG JIT O2. Runtime entries are winner/AmpliCol wall-time multipliers. For the LC all-flow workload, the generation multiplier compares pyAmpliCol process generation with AmpliCol direct-evaluation setup; it is a setup-cost indicator across different boundaries, not a like-for-like compiler benchmark. Summary mode counts use the order A/B/C.

5.2 Built-in SM best measured mode versus AmpliCol: NLC

Built-in SM best measured mode versus AmpliCol NLC (block 1 of 2)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	33.3 (x0.195) (A) 0.227 (x1.14)	26.3 (x0.248) (A) 0.662 (x0.983)	27.6 (x0.238) (A) 3.5 (x0.822)
2 $u\bar{d} \to W^+ + (n-1)g$	32 (x0.204) (A) 0.138 (x1.39)	27.6 (x0.25) (A) 0.375 (x1.19)	25.7 (x0.255) (A) 1.95 (x0.837)
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	29.3 (x0.234) (A) 0.467 (x0.653)	27.2 (x0.24) (A) 1.09 (x0.564)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	29.2 (x0.224) (A) 0.249 (x0.596)	28.6 (x0.23) (A) 0.507 (x0.534)
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	28.9 (x0.226) (A) 0.938 (x0.959)	27.4 (x0.237) (A) 2.84 (x0.992)
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	29.4 (x0.238) (A) 1.7 (x0.916)	35.7 (x0.188) (A) 16.4 (x0.7)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	28.7 (x0.227) (A) 1.16 (x0.602)	29.8 (x0.222) (A) 6.19 (x0.723)
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	29.5 (x0.227) (A) 7.99 (x0.815)	32.4 (x0.224) (A) 121 (x0.448)
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	36.4 (x0.187) (A) 5.65 (x1.04)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	32.7 (x0.199) [A:2/B:0/C:0]	28.6 (x0.234) [A:8/B:0/C:0]	30.1 (x0.222) [A:9/B:0/C:0]
summary: wall	0.183 (x1.23)	1.69 (x0.828)	17.6 (x0.53)

The candidate is selected independently in each cell and workload by the smallest validated wall time. Generation multipliers identify that runtime winner: (A) recurrence JIT O2, (B) compiled JIT O3, and (C) eager-DAG JIT O2. Runtime entries are winner/AmpliCol wall-time multipliers. Summary mode counts use the order A/B/C.

Built-in SM best measured mode versus AmpliCol NLC (block 2 of 2)

ID base process	n=4	n=5
1 $d\bar{d} \to Z + (n-1)g$	46 (x0.169) (A) 31.3 (x0.694)	52.2 (x0.25) (A) 368 (x0.706)
2 $u\bar{d} \to W^+ + (n-1)g$	47.7 (x0.148) (A) 18.4 (x0.687)	46.8 (x0.215) (A) 232 (x0.674)
3 $d\bar{d} \to e^+e^- + (n-2)g$	43.3 (x0.175) (A) 5.46 (x0.55)	56.2 (x0.146) (A) 45 (x0.412)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	56.2 (x0.157) (A) 2.47 (x0.378)	47.4 (x0.185) (A) 22.4 (x0.34)
5 $d\bar{d} \to ZZ + (n-2)g$	59.6 (x0.169) (A) 15.3 (x0.757)	55.7 (x0.179) (A) 118 (x0.977)
6 $gg \to t\bar{t} + (n-2)g$	82.2 (x0.131) (A) 217 (x0.616)	103 (x0.465) (A) 3.52e+03 (x1.62)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	50.3 (x0.166) (A) 57.6 (x0.54)	93.4 (x0.152) (A) 714 (x0.822)
8 $gg \to gg + (n-2)g$	70.2 (x0.315) (A) 2.03e+03 (x0.761)	70.4 (x8.76) (A) 3.67e+04 (x2.38)
9 $d\bar{d} \to ZZZ + (n-3)g$	37.1 (x0.215) (A) 15.1 (x0.776)	43 (x0.179) (A) 74.9 (x0.693)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	118 (x0.0754) (A) 3.93 (x0.735)	152 (x0.0535) (A) 8.85 (x0.722)
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	86 (x0.0987) (A) 11.7 (x0.628)	150 (x0.0537) (A) 65.2 (x0.634)
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	124 (x0.0717) (A) 7.33 (x0.418)	152 (x0.0492) (A) 15.6 (x0.395)
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	N/A N/A N/A N/A	N/A N/A N/A N/A
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE
summary: generation	68.4 (x0.142) [A:12/B:0/C:0]	85.2 (x0.743) [A:12/B:0/C:0]
summary: wall	201 (x0.738)	3.49e+03 (x2.25)

The candidate is selected independently in each cell and workload by the smallest validated wall time. Generation multipliers identify that runtime winner: (A) recurrence JIT O2, (B) compiled JIT O3, and (C) eager-DAG JIT O2. Runtime entries are winner/AmpliCol wall-time multipliers. Summary mode counts use the order A/B/C.

5.3 Built-in SM best measured mode versus AmpliCol: full-colour

Built-in SM best measured mode versus AmpliCol full-colour (block 1 of 2)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	32 (x0.204) (A) 0.271 (x0.978)	27.5 (x0.244) (A) 0.743 (x0.965)	25.8 (x0.258) (A) 3.9 (x0.744)
2 $u\bar{d} \to W^+ + (n-1)g$	32.4 (x0.243) (A) 0.157 (x1.26)	25.7 (x0.255) (A) 0.413 (x1.07)	27.2 (x0.24) (A) 2.16 (x0.883)
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	27.7 (x0.238) (A) 0.522 (x0.595)	29.3 (x0.223) (A) 1.21 (x0.511)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	27 (x0.244) (A) 0.273 (x0.564)	28.5 (x0.234) (A) 0.561 (x0.487)
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	27.5 (x0.242) (A) 1.05 (x0.953)	27.7 (x0.238) (A) 3.09 (x0.895)
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	27.5 (x0.241) (A) 1.97 (x0.777)	35 (x0.196) (A) 19.5 (x0.674)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	29.2 (x0.233) (A) 1.49 (x0.467)	30.5 (x0.216) (A) 7.36 (x0.603)
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	28 (x0.241) (A) 9.41 (x0.633)	32.5 (x0.226) (A) 145 (x0.392)
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	37.3 (x0.184) (A) 6.02 (x0.966)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	32.2 (x0.224) [A:2/B:0/C:0]	27.5 (x0.242) [A:8/B:0/C:0]	30.4 (x0.221) [A:9/B:0/C:0]
summary: wall	0.214 (x1.08)	1.98 (x0.681)	21 (x0.47)

The candidate is selected independently in each cell and workload by the smallest validated wall time. Generation multipliers identify that runtime winner: (A) recurrence JIT O2, (B) compiled JIT O3, and (C) eager-DAG JIT O2. Runtime entries are winner/AmpliCol wall-time multipliers. Summary mode counts use the order A/B/C.

Built-in SM best measured mode versus AmpliCol full-colour (block 2 of 2)

ID	base process	n=4	n=5
1	$d\bar{d} \to Z + (n-1)g$	68.5 (x0.119) (A) 36.3 (x0.81)	79 (x0.178) (A) 471 (x0.646)
2	$u\bar{d} \to W^+ + (n-1)g$	77.3 (x0.115) (A) 21 (x0.734)	33.9 (x0.295) (A) 283 (x0.486)
3	$d\bar{d} \to e^+e^- + (n-2)g$	72.8 (x0.114) (A) 6.07 (x0.504)	77.7 (x0.121) (A) 50.9 (x0.468)
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	64 (x0.139) (A) 2.76 (x0.399)	40.8 (x0.219) (A) 25.9 (x0.366)
5	$d\bar{d} \to ZZ + (n-2)g$	70.6 (x0.125) (A) 16.7 (x0.769)	81.7 (x0.12) (A) 133 (x0.83)
6	$gg \to t\bar{t} + (n-2)g$	96.7 (x0.123) (A) 284 (x0.492)	141 (x0.324) (A) 6.58e+03 (x1.22)
7	$d\bar{d} \to t\bar{t} + (n-2)g$	85.5 (x0.118) (A) 73.5 (x0.571)	99.1 (x0.164) (A) 1.1e+03 (x0.561)
8	$gg \to gg + (n-2)g$	79.7 (x0.308) (A) 3.07e+03 (x0.762)	92 (x6.44) (A) 9.73e+04 (x1.16)
9	$d\bar{d} \to ZZZ + (n-3)g$	37 (x0.219) (A) 15.9 (x0.769)	42.7 (x0.202) (A) 79.6 (x0.754)
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	118 (x0.0815) (A) 4.09 (x0.989)	143 (x0.0606) (A) 9.19 (x0.787)
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	94.9 (x0.0913) (A) 12.7 (x0.567)	149 (x0.0656) (A) 71 (x0.52)
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	118 (x0.0687) (A) 7.59 (x0.425)	145 (x0.0646) (A) 16.1 (x0.485)
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	N/A N/A N/A N/A	N/A N/A N/A N/A
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE
summary: generation		81.9 (x0.126) [A:12/B:0/C:0]	93.8 (x0.66) [A:12/B:0/C:0]
summary: wall		296 (x0.735)	8.84e+03 (x1.15)

The candidate is selected independently in each cell and workload by the smallest validated wall time. Generation multipliers identify that runtime winner: (A) recurrence JIT O2, (B) compiled JIT O3, and (C) eager-DAG JIT O2. Runtime entries are winner/AmpliCol wall-time multipliers. Summary mode counts use the order A/B/C.

The detailed tables are grouped by execution mode, model source, and colour treatment.

5.4 Built-in SM recurrence versus AmpliCol LC

Built-in SM recurrence versus AmpliCol LC (block 1 of 3)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	31.9 / 0.000361 (x0.204) C 6.51; N.C. 0.0565 / 0.125 (x4.34|x3.05) (x1.08|x0.151)	25.6 / 0.000914 (x0.261) C 6.61; N.C. 0.366 / 0.288 (x1.87|x1.55) (x0.969|x0.287)	27.4 / 0.00149 (x0.246) C 6.61; N.C. 1.23 / 0.834 (x1.2|x0.997) (x0.846|x0.475)
2 $u\bar{d} \to W^+ + (n-1)g$	31.6 / 0.000382 (x0.209) C 6.54; N.C. 0.0321 / 0.124 (x6.02|x4.1) (x1.1|x0.168)	26.2 / 0.000411 (x0.252) C 7.23; N.C. 0.22 / 0.288 (x2|x1.46) (x0.979|x0.296)	27 / 0.000656 (x0.244) C 6.57; N.C. 0.702 / 0.749 (x1.45|x1.15) (x0.889|x0.475)
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	26.7 / 0.000927 (x0.244) C 6.68; N.C. 0.182 / 0.262 (x1.7|x1.17) (x0.861|x0.337)	27.8 / 0.000602 (x0.24) C 6.68; N.C. 0.445 / 0.434 (x1.44|x1.1) (x0.86|x0.384)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	29.1 / 0.000701 (x0.232) C 7.3; N.C. 0.0673 / 0.206 (x2.45|x1.14) (x1.72|x0.254)	27.1 / 0.00133 (x0.242) C 6.54; N.C. 0.139 / 0.361 (x2.2|x1.19) (x0.917|x0.343)
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	29.2 / 0.000861 (x0.228) C 6.6; N.C. 0.47 / 0.298 (x1.83|x1.51) (x0.951|x0.265)	28.1 / 0.0014 (x0.24) C 6.7; N.C. 2.09 / 0.608 (x1.25|x1.1) (x0.881|x0.381)
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	27.3 / 0.000944 (x0.256) C 7.19; N.C. 0.32 / 0.413 (x2.7|x2.23) (x0.956|x0.387)	33.1 / 0.00174 (x0.203) C 6.69; N.C. 1.94 / 1.63 (x2.37|x2.24) (x0.808|x0.576)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	28.7 / 0.000418 (x0.231) C 7.12; N.C. 0.144 / 0.286 (x4.12|x3.05) (x0.959|x0.263)	30.7 / 0.000668 (x0.217) C 6.54; N.C. 0.63 / 0.945 (x4.82|x4.3) (x0.859|x0.492)
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	31.2 / 0.000446 (x0.21) C 7.15; N.C. 0.459 / 0.608 (x4.14|x3.75) (x1.36|x0.936)	29.7 / 0.00094 (x0.229) C 6.66; N.C. 2.4 / 3.77 (x3.37|x3.25) (x1.32|x1.2)
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	34.9 / 0.000633 (x0.2) C 7.52; N.C. 4.59 / 0.83 (x1.03|x0.952) (x0.85|x0.442)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	31.8 (x0.207) / 0.000371 C 6.53; N.C.	28 (x0.238) / 0.000703 C 6.99; N.C.	29.5 (x0.227) / 0.00105 C 6.72; N.C.
summary: wall	0.0443 (x4.95) / 0.124 (x1.09)	0.279 (x2.61) / 0.331 (x1.1)	1.57 (x1.87) / 1.13 (x1.02)

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time marked n.c. (not comparable); no generation ratio is formed. Runtime comparisons use native wall time; a separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

Built-in SM recurrence versus AmpliCol LC (block 2 of 3)

ID base process	n=4	n=5	n=6
1 $d\bar{d} \to Z + (n-1)g$	64.7 / 0.00227 (x0.154) C 9.36; N.C. 4.16 / 2.93 (x1.09|x0.983) (x0.851|x0.701)	38.8 / 0.0149 (x0.195) C 9.14; N.C. 12.4 / 15.2 (x0.865|x0.843) (x0.579|x0.538)	151 / 0.229 (x0.0803) C 10.3; N.C. 157 / 97.2 (x0.227|x0.228) (x0.754|x0.743)
2 $u\bar{d} \to W^+ + (n-1)g$	46 / 0.00502 (x0.152) C 8.2; N.C. 2.52 / 2.97 (x1.26|x1.15) (x0.804|x0.635)	51 / 0.0149 (x0.146) C 8.95; N.C. 8.26 / 15.3 (x1.34|x1.36) (x0.812|x0.712)	93.1 / 0.137 (x0.122) C 11; N.C. 24.2 / 96.3 (x1.85|x0.797) (x0.826|x0.763)
3 $d\bar{d} \to e^+e^- + (n-2)g$	76.9 / 0.00193 (x0.101) C 11.2; N.C. 1.25 / 0.968 (x1.34|x1.14) (x0.917|x0.524)	37.2 / 0.0116 (x0.229) C 8.46; N.C. 3.83 / 3.52 (x1.06|x0.941) (x0.896|x0.652)	155 / 0.0965 (x0.0851) C 12; N.C. 52.3 / 16.8 (x0.319|x0.28) (x1.13|x0.934)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	49.1 / 0.00413 (x0.142) C 9.02; N.C. 0.417 / 0.821 (x1.69|x1.18) (x1.12|x0.6)	42.5 / 0.0113 (x0.186) C 9.46; N.C. 1.69 / 3.08 (x1.54|x1.19) (x0.891|x0.671)	92.1 / 0.0951 (x0.103) C 10.6; N.C. 6.15 / 15.1 (x2.92|x1.34) (x0.793|x0.671)
5 $d\bar{d} \to ZZ + (n-2)g$	81 / 0.00209 (x0.112) C 9.1; N.C. 6.29 / 1.65 (x1.05|x0.999) (x0.816|x0.534)	41.7 / 0.0128 (x0.226) C 9.11; N.C. 16.8 / 6.31 (x0.937|x0.934) (x0.797|x0.636)	161 / 0.15 (x0.0757) C 9.77; N.C. 98.3 / 30.7 (x0.442|x0.434) (x0.596|x0.546)
6 $gg \to t\bar{t} + (n-2)g$	78 / 0.0032 (x0.106) C 9.21; N.C. 6.87 / 8.4 (x5.04|x5.37) (x1.07|x0.983)	141 / 0.0261 (x0.077) C 10.4; N.C. 21.1 / 55 (x7.9|x8.18) (x0.723|x0.767)	389 / 0.292 (x0.187) C 20.5; N.C. 58.4 / 480 (x11.4|x11.4) (x0.607|x0.604)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	92.2 / 0.00225 (x0.109) C 9.4; N.C. 2.41 / 4.1 (x6.41|x5.81) (x1.09|x0.871)	59.5 / 0.0145 (x0.166) C 8.72; N.C. 8.12 / 23.5 (x8.59|x8.93) (x0.564|x0.538)	246 / 0.139 (x0.099) C 13.6; N.C. 90.8 / 176 (x6.5|x3.58) (x0.517|x0.5)
8 $gg \to gg + (n-2)g$	58.9 / 0.00509 (x0.167) C 7.98; N.C. 9.23 / 26.1 (x4.18|x4.14) (x0.877|x0.856)	56.7 / 0.0716 (x0.272) C 13.5; N.C. 29.4 / 219 (x4.75|x4.69) (x0.956|x0.904)	217 / 3.08 (x0.847) C 165; N.C. 356 / 3.88e+03 (x1.26|x1.24) (x0.873|x0.866)
9 $d\bar{d} \to ZZZ + (n-3)g$	37.2 / 0.00348 (x0.227) C 7.25; N.C. 12.9 / 1.56 (x0.836|x0.789) (x0.845|x0.501)	43.3 / 0.0112 (x0.174) C 7.67; N.C. 33.2 / 3.94 (x0.779|x0.742) (x0.718|x0.516)	57.8 / 0.0946 (x0.131) C 7.79; N.C. 84.9 / 14.5 (x0.735|x0.709) (x0.636|x0.54)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	163 / 0.00196 (x0.0506) C 7.88; N.C. 3.78 / 1.79 (x0.853|x0.773) (x1.26|x0.965)	103 / 0.011 (x0.0714) C 9.63; N.C. 7.87 / 2.66 (x0.792|x0.674) (x1.02|x0.785)	297 / 0.0889 (x0.0304) C 10.5; N.C. 17.1 / 4.91 (x0.668|x0.625) (x0.858|x0.668)
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	82.2 / 0.00146 (x0.105) C 8.81; N.C. 0.462 / 2.04 (x8.36|x7.42) (x1.66|x1.24)	46 / 0.0101 (x0.185) C 8.94; N.C. 1.11 / 9.62 (x38.8|x36.5) (x0.852|x0.769)	223 / 0.0996 (x0.0962) C 11.9; N.C. 2.97 / 53.3 (x257|x119) (x1.18|x1.12)
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	N/A / N/A N/A N/A N/A / N/A N/A N/A
summary: generation	75.4 (x0.114) / 0.00299 C 8.85; N.C.	60.1 (x0.152) / 0.0191 C 9.46; N.C.	189 (x0.181) / 0.409 C 25.7; N.C.
summary: wall	4.57 (x2.45) / 4.85 (x0.964)	13.1 (x3.45) / 32.4 (x0.863)	86.1 (x2.85) / 442 (x0.832)

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time marked n.c. (not comparable); no generation ratio is formed. Runtime comparisons use native wall time; a separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

Built-in SM recurrence versus AmpliCol LC (block 3 of 3)

ID base process	n=7	n=8	n=9
1 $d\bar{d} \to Z + (n-1)g$	29.2 / 1.45 (x0.4) C 39.3; N.C. 91.9 / 853 (x0.691|x0.665) (x0.504|x0.507)	63.5 / N/A (x1.32) N/A 234 / N/A (x0.706|x0.648) N/A	62.5 / 511 (x18.1) N/A 568 / 2.7e+05 (x0.728|x0.706) N/A
2 $u\bar{d} \to W^+ + (n-1)g$	46.4 / 1.46 (x0.258) C 45.9; N.C. 65.6 / 916 (x0.732|x0.697) (x0.553|x0.587)	59.9 / N/A (x1.09) N/A 192 / N/A (x0.618|x0.577) N/A	45.7 / 511 (x18.1) N/A 435 / 2.19e+05 (x0.678|x0.659) N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	62.4 / 0.962 (x0.142) C 10.4; N.C. 31.7 / 105 (x0.86|x0.821) (x0.573|x0.52)	58.8 / N/A (x0.19) N/A 87.6 / N/A (x0.68|x0.624) N/A	48.5 / N/A (x1.69) N/A 208 / N/A (x0.717|x0.666) N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	61.5 / 0.932 (x0.158) C 12.3; N.C. 19.2 / 96.1 (x0.819|x0.768) (x0.604|x0.545)	67.5 / N/A (x0.16) N/A 54.4 / N/A (x1.44|x1.29) N/A	35.9 / N/A (x1.67) N/A 143 / N/A (x0.721|x0.642) N/A
5 $d\bar{d} \to ZZ + (n-2)g$	73.7 / 1.02 (x0.144) C 23.3; N.C. 111 / 195 (x0.858|x0.825) (x0.511|x0.486)	129 / N/A (x0.241) N/A 271 / N/A (x0.901|x0.905) N/A	141 / N/A (x1.42) N/A 648 / N/A (x0.782|x0.709) N/A
6 $gg \to t\bar{t} + (n-2)g$	1.25e+03 / 5.84 (x1.65) C 408; N.C. 154 / 1.23e+04 (x41.4|x40.9) (x0.624|x0.621)	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
7 $d\bar{d} \to t\bar{t} + (n-2)g$	136 / 1.58 (x1.2) C 70; N.C. 65.5 / 2.02e+03 (x19|x18.4) (x0.511|x0.508)	491 / 28.7 (x7.74) N/A 170 / 2.41e+04 (x34.5|x34.3) N/A	4.23e+03 / N/A N/A N/A 582 / N/A N/A N/A
8 $gg \to gg + (n-2)g$	136 / 234 (x32.4) N/A 212 / 7.03e+04 (x7.68|x7.61) N/A	397 / N/A N/A N/A 539 / N/A N/A N/A	NOT APPLICABLE
9 $d\bar{d} \to ZZZ + (n-3)g$	102 / 0.946 (x0.0866) C 12.4; N.C. 203 / 66.6 (x0.75|x0.735) (x0.472|x0.434)	268 / 10.9 (x0.0525) C 62.5; N.C. 476 / 414 (x0.761|x0.749) (x0.427|x0.428)	743 / 124 (x0.0872) C 1.18e+03; N.C. 1.25e+03 / 7.52e+03 (x0.727|x0.715) (x0.264|x0.266)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	178 / N/A N/A N/A 61.3 / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	154 / 0.872 (x0.0685) C 9.87; N.C. 39.3 / 15.4 (x0.581|x0.533) (x0.717|x0.601)	193 / N/A (x0.0422) N/A 92.2 / N/A (x0.526|x0.457) N/A	199 / N/A (x0.0575) N/A 210 / N/A (x1.22|x1.01) N/A
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	134 / 1.34 (x1.17) C 23.4; N.C. 7.91 / 375 (x513|x503) (x0.761|x0.777)	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
summary: generation	198 (x3.14) / 1.64 C 65.5; N.C.	166 (x3.02) / 10.9 C 62.5; N.C.	182 (x1.86) / 124 C 1.18e+03; N.C.
summary: wall	91 (x13.7) / 1.7e+03 (x0.602)	197 (x4.41) / 414 (x0.427)	494 (x0.76) / 7.52e+03 (x0.264)

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time marked n.c. (not comparable); no generation ratio is formed. Runtime comparisons use native wall time; a separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

5.5 Built-in SM recurrence versus AmpliCol NLC

Built-in SM recurrence versus AmpliCol NLC (block 1 of 2)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	33.3 (x0.195) 0.227 (x1.14|x0.737)	26.3 (x0.248) 0.662 (x0.983|x0.72)	27.6 (x0.238) 3.5 (x0.822|x0.597)
2 $u\bar{d} \to W^+ + (n-1)g$	32 (x0.204) 0.138 (x1.39|x0.921)	27.6 (x0.25) 0.375 (x1.19|x0.86)	25.7 (x0.255) 1.95 (x0.837|x0.605)
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	29.3 (x0.234) 0.467 (x0.653|x0.457)	27.2 (x0.24) 1.09 (x0.564|x0.432)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	29.2 (x0.224) 0.249 (x0.596|x0.315)	28.6 (x0.23) 0.507 (x0.534|x0.321)
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	28.9 (x0.226) 0.938 (x0.959|x0.721)	27.4 (x0.237) 2.84 (x0.992|x0.834)
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	29.4 (x0.238) 1.7 (x0.916|x0.622)	35.7 (x0.188) 16.4 (x0.7|x0.431)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	28.7 (x0.227) 1.16 (x0.602|x0.346)	29.8 (x0.222) 6.19 (x0.723|x0.517)
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	29.5 (x0.227) 7.99 (x0.815|x0.652)	32.4 (x0.224) 121 (x0.448|x0.351)
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	36.4 (x0.187) 5.65 (x1.04|x0.951)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	32.7 (x0.199)	28.6 (x0.234)	30.1 (x0.222)
summary: wall	0.183 (x1.23)	1.69 (x0.828)	17.6 (x0.53)

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

Built-in SM recurrence versus AmpliCol NLC (block 2 of 2)

ID base process	n=4	n=5
1 $d\bar{d} \to Z + (n-1)g$	46 (x0.169) 31.3 (x0.694|x0.48)	52.2 (x0.25) 368 (x0.706|x0.488)
2 $u\bar{d} \to W^+ + (n-1)g$	47.7 (x0.148) 18.4 (x0.687|x0.495)	46.8 (x0.215) 232 (x0.674|x0.485)
3 $d\bar{d} \to e^+e^- + (n-2)g$	43.3 (x0.175) 5.46 (x0.55|x0.443)	56.2 (x0.146) 45 (x0.412|x0.296)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	56.2 (x0.157) 2.47 (x0.378|x0.26)	47.4 (x0.185) 22.4 (x0.34|x0.288)
5 $d\bar{d} \to ZZ + (n-2)g$	59.6 (x0.169) 15.3 (x0.757|x0.627)	55.7 (x0.179) 118 (x0.977|x0.773)
6 $gg \to t\bar{t} + (n-2)g$	82.2 (x0.131) 217 (x0.616|x0.385)	103 (x0.465) 3.52e+03 (x1.62|x1.14)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	50.3 (x0.166) 57.6 (x0.54|x0.408)	93.4 (x0.152) 714 (x0.822|x0.571)
8 $gg \to gg + (n-2)g$	70.2 (x0.315) 2.03e+03 (x0.761|x0.475)	70.4 (x8.76) 3.67e+04 (x2.38|x1.88)
9 $d\bar{d} \to ZZZ + (n-3)g$	37.1 (x0.215) 15.1 (x0.776|x0.698)	43 (x0.179) 74.9 (x0.693|x0.608)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	118 (x0.0754) 3.93 (x0.735|x0.687)	152 (x0.0535) 8.85 (x0.722|x0.637)
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	86 (x0.0987) 11.7 (x0.628|x0.517)	150 (x0.0537) 65.2 (x0.634|x0.535)
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	124 (x0.0717) 7.33 (x0.418|x0.379)	152 (x0.0492) 15.6 (x0.395|x0.337)
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	N/A N/A N/A N/A	N/A N/A N/A N/A
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE
summary: generation	68.4 (x0.142)	85.2 (x0.743)
summary: wall	201 (x0.738)	3.49e+03 (x2.25)

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

5.6 Built-in SM recurrence versus AmpliCol full-colour

Built-in SM recurrence versus AmpliCol full-colour (block 1 of 2)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	32 (x0.204) 0.271 (x0.978|x0.654)	27.5 (x0.244) 0.743 (x0.965|x0.73)	25.8 (x0.258) 3.9 (x0.744|x0.535)
2 $u\bar{d} \to W^+ + (n-1)g$	32.4 (x0.243) 0.157 (x1.26|x0.865)	25.7 (x0.255) 0.413 (x1.07|x0.776)	27.2 (x0.24) 2.16 (x0.883|x0.68)
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	27.7 (x0.238) 0.522 (x0.595|x0.414)	29.3 (x0.223) 1.21 (x0.511|x0.377)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	27 (x0.244) 0.273 (x0.564|x0.299)	28.5 (x0.234) 0.561 (x0.487|x0.293)
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	27.5 (x0.242) 1.05 (x0.953|x0.771)	27.7 (x0.238) 3.09 (x0.895|x0.751)
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	27.5 (x0.241) 1.97 (x0.777|x0.513)	35 (x0.196) 19.5 (x0.674|x0.362)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	29.2 (x0.233) 1.49 (x0.467|x0.275)	30.5 (x0.216) 7.36 (x0.603|x0.425)
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	28 (x0.241) 9.41 (x0.633|x0.48)	32.5 (x0.226) 145 (x0.392|x0.295)
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	37.3 (x0.184) 6.02 (x0.966|x0.878)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	32.2 (x0.224)	27.5 (x0.242)	30.4 (x0.221)
summary: wall	0.214 (x1.08)	1.98 (x0.681)	21 (x0.47)

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

Built-in SM recurrence versus AmpliCol full-colour (block 2 of 2)

ID base process	n=4	n=5
1 $d\bar{d} \to Z + (n-1)g$	68.5 (x0.119) 36.3 (x0.81|x0.487)	79 (x0.178) 471 (x0.646|x0.464)
2 $u\bar{d} \to W^+ + (n-1)g$	77.3 (x0.115) 21 (x0.734|x0.491)	33.9 (x0.295) 283 (x0.486|x0.361)
3 $d\bar{d} \to e^+e^- + (n-2)g$	72.8 (x0.114) 6.07 (x0.504|x0.415)	77.7 (x0.121) 50.9 (x0.468|x0.331)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	64 (x0.139) 2.76 (x0.399|x0.289)	40.8 (x0.219) 25.9 (x0.366|x0.3)
5 $d\bar{d} \to ZZ + (n-2)g$	70.6 (x0.125) 16.7 (x0.769|x0.604)	81.7 (x0.12) 133 (x0.83|x0.606)
6 $gg \to t\bar{t} + (n-2)g$	96.7 (x0.123) 284 (x0.492|x0.3)	141 (x0.324) 6.58e+03 (x1.22|x0.734)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	85.5 (x0.118) 73.5 (x0.571|x0.414)	99.1 (x0.164) 1.1e+03 (x0.561|x0.366)
8 $gg \to gg + (n-2)g$	79.7 (x0.308) 3.07e+03 (x0.762|x0.382)	92 (x6.44) 9.73e+04 (x1.16|x0.735)
9 $d\bar{d} \to ZZZ + (n-3)g$	37 (x0.219) 15.9 (x0.769|x0.685)	42.7 (x0.202) 79.6 (x0.754|x0.672)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	118 (x0.0815) 4.09 (x0.989|x0.846)	143 (x0.0606) 9.19 (x0.787|x0.739)
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	94.9 (x0.0913) 12.7 (x0.567|x0.501)	149 (x0.0656) 71 (x0.52|x0.46)
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	118 (x0.0687) 7.59 (x0.425|x0.377)	145 (x0.0646) 16.1 (x0.485|x0.463)
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	N/A N/A N/A N/A	N/A N/A N/A N/A
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE
summary: generation	81.9 (x0.126)	93.8 (x0.66)
summary: wall	296 (x0.735)	8.84e+03 (x1.15)

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

5.7 UFO-SM recurrence versus AmpliCol LC

UFO-SM recurrence versus AmpliCol LC (block 1 of 3)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	31.9 / 0.000361 (x0.325) C 10.2; N.C. 0.0565 / 0.125 (x4.39|x3.05) (x1.11|x0.178)	25.6 / 0.000914 (x0.411) C 10.4; N.C. 0.366 / 0.288 (x1.73|x1.36) (x1.04|x0.339)	27.4 / 0.00149 (x0.651) C 13.7; N.C. 1.23 / 0.834 (x1.92|x1.61) (x0.942|x0.549)
2 $u\bar{d} \to W^+ + (n-1)g$	31.6 / 0.000382 (x0.324) C 10.3; N.C. 0.0321 / 0.124 (x6.01|x3.88) (x1.05|x0.117)	26.2 / 0.000411 (x0.397) C 10.3; N.C. 0.22 / 0.288 (x2.05|x1.5) (x0.981|x0.286)	27 / 0.000656 (x0.63) C 12.6; N.C. 0.702 / 0.749 (x1.88|x1.44) (x0.875|x0.46)
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	26.7 / 0.000927 (x0.413) C 10.4; N.C. 0.182 / 0.262 (x1.72|x1.21) (x0.904|x0.373)	27.8 / 0.000602 (x0.542) C 13.2; N.C. 0.445 / 0.434 (x2.38|x1.77) (x0.935|x0.429)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	29.1 / 0.000701 (x0.364) C 10.1; N.C. 0.0673 / 0.206 (x5.07|x1.16) (x0.898|x0.214)	27.1 / 0.00133 (x0.56) C 12.5; N.C. 0.139 / 0.361 (x2.66|x1.49) (x0.955|x0.311)
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	29.2 / 0.000861 (x0.36) C 10.4; N.C. 0.47 / 0.298 (x1.86|x1.54) (x1.04|x0.35)	28.1 / 0.0014 (x0.578) C 12.2; N.C. 2.09 / 0.608 (x1.77|x1.51) (x1.13|x0.551)
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	27.3 / 0.000944 (x0.393) C 10.3; N.C. 0.32 / 0.413 (x2.56|x2.08) (x1.2|x0.624)	33.1 / 0.00174 (x0.5) C 12.3; N.C. 1.94 / 1.63 (x3.48|x3.21) (x1.08|x0.849)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	28.7 / 0.000418 (x0.383) C 10.6; N.C. 0.144 / 0.286 (x3.88|x2.89) (x0.992|x0.293)	30.7 / 0.000668 (x0.492) C 11.9; N.C. 0.63 / 0.945 (x8.39|x7.3) (x0.967|x0.569)
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	31.2 / 0.000446 (x0.345) C 10.5; N.C. 0.459 / 0.608 (x4.33|x3.87) (x1.18|x0.74)	29.7 / 0.00094 (x0.486) C 12.7; N.C. 2.4 / 3.77 (x4.77|x4.79) (x1.98|x1.89)
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	34.9 / 0.000633 (x0.32) C 11.4; N.C. 4.59 / 0.83 (x1.09|x0.993) (x0.879|x0.454)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	31.8 (x0.325) / 0.000371 C 10.3; N.C.	28 (x0.382) / 0.000703 C 10.4; N.C.	29.5 (x0.522) / 0.00105 C 12.5; N.C.
summary: wall	0.0443 (x4.98) / 0.124 (x1.08)	0.279 (x2.68) / 0.331 (x1.06)	1.57 (x2.63) / 1.13 (x1.35)

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time marked n.c. (not comparable); no generation ratio is formed. Runtime comparisons use native wall time; a separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

UFO-SM recurrence versus AmpliCol LC (block 2 of 3)

<table>
<tr><th>ID base process</th><th>n=4</th><th>n=5</th><th>n=6</th></tr>
<tr><td>1 $d\bar{d} \to Z + (n-1)g$</td><td>64.7 / 0.00227 (x0.218) C 14.1; n.c.
4.16 / 2.93 (x1.05|x0.942) (x1.2|x1.01)</td><td>38.8 / 0.0149 (x0.347) C 40.9; n.c.
12.4 / 15.2 (x0.856|x0.797) (x2.57|x1.99)</td><td>151 / 0.229 (x0.0873) C 17.8; n.c.
157 / 97.2 (x0.188|x0.184) (x1.04|x0.953)</td></tr>
<tr><td>2 $u\bar{d} \to W^+ + (n-1)g$</td><td>46 / 0.00502 (x0.253) C 13.3; n.c.
2.52 / 2.97 (x1.54|x1.42) (x1.3|x0.995)</td><td>51 / 0.0149 (x0.268) C 45.2; n.c.
8.26 / 15.3 (x1.54|x1.47) (x2.25|x1.94)</td><td>93.1 / 0.137 (x0.126) C 15.8; n.c.
24.2 / 96.3 (x0.708|x0.673) (x1.04|x0.931)</td></tr>
<tr><td>3 $d\bar{d} \to e^+e^- + (n-2)g$</td><td>76.9 / 0.00193 (x0.167) C 16; n.c.
1.25 / 0.968 (x1.31|x1.05) (x1.02|x0.621)</td><td>37.2 / 0.0116 (x0.352) C 40; n.c.
3.83 / 3.52 (x1.24|x1.04) (x2.86|x2.45)</td><td>155 / 0.0965 (x0.0987) C 15.9; n.c.
52.3 / 16.8 (x0.181|x0.167) (x0.626|x0.572)</td></tr>
<tr><td>4 $u\bar{d} \to e^+\nu_e + (n-2)g$</td><td>49.1 / 0.00413 (x0.263) C 13.8; n.c.
0.417 / 0.821 (x1.73|x1.21) (x1.22|x0.652)</td><td>42.5 / 0.0113 (x0.32) C 42.2; n.c.
1.69 / 3.08 (x1.77|x1.54) (x2.91|x2.22)</td><td>92.1 / 0.0951 (x0.159) C 15.5; n.c.
6.15 / 15.1 (x1.45|x1.35) (x0.635|x0.556)</td></tr>
<tr><td>5 $d\bar{d} \to ZZ + (n-2)g$</td><td>81 / 0.00209 (x0.168) C 13.8; n.c.
6.29 / 1.65 (x1.31|x1.25) (x1.05|x0.676)</td><td>41.7 / 0.0128 (x0.359) C 41.3; n.c.
16.8 / 6.31 (x0.989|x0.907) (x2.38|x2.47)</td><td>161 / 0.15 (x0.076) C 15; n.c.
98.3 / 30.7 (x0.375|x0.362) (x1.1|x0.976)</td></tr>
<tr><td>6 $gg \to t\bar{t} + (n-2)g$</td><td>78 / 0.0032 (x0.174) C 14.8; n.c.
6.87 / 8.4 (x4.73|x4.51) (x1.56|x1.39)</td><td>141 / 0.0261 (x0.119) C 49.5; n.c.
21.1 / 55 (x8.5|x7.85) (x2.26|x2.45)</td><td>389 / 0.292 (x0.16) C 29.1; n.c.
58.4 / 480 (x11.6|x11.6) (x0.569|x0.556)</td></tr>
<tr><td>7 $d\bar{d} \to t\bar{t} + (n-2)g$</td><td>92.2 / 0.00225 (x0.139) C 15.4; n.c.
2.41 / 4.1 (x5.9|x5.71) (x1.29|x1.04)</td><td>59.5 / 0.0145 (x0.24) C 42; n.c.
8.12 / 23.5 (x8.16|x8.05) (x2.02|x1.81)</td><td>246 / 0.139 (x0.0864) C 19; n.c.
90.8 / 176 (x3.03|x3.03) (x0.593|x0.567)</td></tr>
<tr><td>8 $gg \to gg + (n-2)g$</td><td>58.9 / 0.00509 (x0.264) C 13.7; n.c.
9.23 / 26.1 (x3.78|x3.74) (x0.836|x0.793)</td><td>56.7 / 0.0716 (x0.408) C 57.4; n.c.
29.4 / 219 (x4.84|x4.77) (x2.39|x2.64)</td><td>217 / 3.08 (x0.831) C 158; n.c.
356 / 3.88e+03 (x1.26|x1.24) (x0.899|x0.89)</td></tr>
<tr><td>9 $d\bar{d} \to ZZZ + (n-3)g$</td><td>37.2 / 0.00348 (x0.317) C 11.1; n.c.
12.9 / 1.56 (x0.896|x0.848) (x1.38|x1.05)</td><td>43.3 / 0.0112 (x0.263) C 11.1; n.c.
33.2 / 3.94 (x0.846|x0.815) (x0.903|x0.684)</td><td>57.8 / 0.0946 (x0.208) C 12; n.c.
84.9 / 14.5 (x0.779|x0.758) (x0.841|x0.73)</td></tr>
<tr><td>10 $d\bar{d} \to e^+e^-ZH + (n-4)g$</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td></tr>
<tr><td>11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td></tr>
<tr><td>12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$</td><td>163 / 0.00196 (x0.0764) C 14.5; n.c.
3.78 / 1.79 (x0.989|x0.881) (x0.804|x0.611)</td><td>103 / 0.011 (x0.132) C 42.6; n.c.
7.87 / 2.66 (x0.677|x0.602) (x2.09|x1.83)</td><td>297 / 0.0889 (x0.0404) C 12.9; n.c.
17.1 / 4.91 (x1.08|x1) (x0.903|x0.756)</td></tr>
<tr><td>13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$</td><td>82.2 / 0.00146 (x0.15) C 13.9; n.c.
0.462 / 2.04 (x8.87|x8.25) (x1.29|x0.926)</td><td>46 / 0.0101 (x0.343) C 44.5; n.c.
1.11 / 9.62 (x35.7|x34.7) (x4.11|x1.41)</td><td>223 / 0.0996 (x0.0937) C 14.9; n.c.
2.97 / 53.3 (x106|x104) (x0.677|x0.632)</td></tr>
<tr><td>14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$</td><td>NOT APPLICABLE</td><td>NOT APPLICABLE</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td></tr>
<tr><td>summary: generation</td><td>75.4 (x0.173) / 0.00299 C 14; n.c.</td><td>60.1 (x0.248) / 0.0191 C 41.5; n.c.</td><td>189 (x0.181) / 0.409 C 29.7; n.c.</td></tr>
<tr><td>summary: wall</td><td>4.57 (x2.38) / 4.85 (x1.08)</td><td>13.1 (x3.54) / 32.4 (x2.38)</td><td>86.1 (x2) / 442 (x0.858)</td></tr>
</table>

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time marked n.c. (not comparable); no generation ratio is formed. Runtime comparisons use native wall time; a separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

UFO-SM recurrence versus AmpliCol LC (block 3 of 3)

<table>
<tr><th>ID base process</th><th>n=7</th><th>n=8</th><th>n=9</th></tr>
<tr><td>1 $d\bar{d} \to Z + (n-1)g$</td><td>29.2 / 1.45 (x0.602) C 46.1; N.C.
91.9 / 853 (x0.719|x0.697) (x0.549|x0.535)</td><td>63.5 / N/A (x1.38) N/A
234 / N/A (x0.684|x0.66) N/A</td><td>62.5 / 511 N/A N/A
568 / 2.7e+05 N/A N/A</td></tr>
<tr><td>2 $u\bar{d} \to W^+ + (n-1)g$</td><td>46.4 / 1.46 (x0.424) C 77; N.C.
65.6 / 916 (x0.727|x0.677) (x0.633|x0.615)</td><td>59.9 / N/A (x1.12) N/A
192 / N/A (x0.602|x0.581) N/A</td><td>45.7 / 511 N/A N/A
435 / 2.19e+05 N/A N/A</td></tr>
<tr><td>3 $d\bar{d} \to e^+e^- + (n-2)g$</td><td>62.4 / 0.962 (x0.21) C 41.2; N.C.
31.7 / 105 (x0.844|x0.874) (x1.03|x1.05)</td><td>58.8 / N/A (x0.361) N/A
87.6 / N/A (x0.848|x0.715) N/A</td><td>48.5 / N/A (x1.67) N/A
208 / N/A (x0.699|x0.682) N/A</td></tr>
<tr><td>4 $u\bar{d} \to e^+\nu_e + (n-2)g$</td><td>61.5 / 0.932 (x0.225) C 27.6; N.C.
19.2 / 96.1 (x0.814|x0.791) (x1.35|x1.21)</td><td>67.5 / N/A (x0.286) N/A
54.4 / N/A (x0.838|x0.838) N/A</td><td>35.9 / N/A (x1.73) N/A
143 / N/A (x0.8|x0.774) N/A</td></tr>
<tr><td>5 $d\bar{d} \to ZZ + (n-2)g$</td><td>73.7 / 1.02 (x0.232) C 63.8; N.C.
111 / 195 (x0.79|x0.753) (x0.523|x0.494)</td><td>129 / N/A (x0.275) N/A
271 / N/A (x0.76|x0.732) N/A</td><td>141 / N/A (x1.49) N/A
648 / N/A (x0.773|x0.72) N/A</td></tr>
<tr><td>6 $gg \to t\bar{t} + (n-2)g$</td><td>1.25e+03 / 5.84 (x1.66) C 447; N.C.
154 / 1.23e+04 (x71.2|x81.7) (x0.672|x0.665)</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td><td>NOT APPLICABLE</td></tr>
<tr><td>7 $d\bar{d} \to t\bar{t} + (n-2)g$</td><td>136 / 1.58 (x1.18) C 94.9; N.C.
65.5 / 2.02e+03 (x19.8|x19.6) (x0.535|x0.53)</td><td>491 / 28.7 N/A N/A
170 / 2.41e+04 N/A N/A</td><td>4.23e+03 / N/A N/A N/A
582 / N/A N/A N/A</td></tr>
<tr><td>8 $gg \to gg + (n-2)g$</td><td>136 / 234 N/A N/A
212 / 7.03e+04 N/A N/A</td><td>397 / N/A N/A N/A
539 / N/A N/A N/A</td><td>NOT APPLICABLE</td></tr>
<tr><td>9 $d\bar{d} \to ZZZ + (n-3)g$</td><td>102 / 0.946 (x0.133) C 17.2; N.C.
203 / 66.6 (x0.771|x0.753) (x0.531|x0.471)</td><td>268 / 10.9 (x0.0737) C 76.7; N.C.
476 / 414 (x0.797|x0.787) (x0.513|x0.49)</td><td>743 / 124 (x0.105) C 1.34e+03; N.C.
1.25e+03 / 7.52e+03 (x0.756|x0.746) (x0.316|x0.311)</td></tr>
<tr><td>10 $d\bar{d} \to e^+e^-ZH + (n-4)g$</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td><td>178 / N/A N/A N/A
61.3 / N/A N/A N/A</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td></tr>
<tr><td>11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td></tr>
<tr><td>12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$</td><td>154 / 0.872 (x0.109) C 29.7; N.C.
39.3 / 15.4 (x0.586|x0.575) (x2.03|x1.71)</td><td>193 / N/A (x0.0835) N/A
92.2 / N/A (x0.596|x0.601) N/A</td><td>199 / N/A (x0.0908) N/A
210 / N/A (x0.623|x0.658) N/A</td></tr>
<tr><td>13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$</td><td>134 / 1.34 (x1.04) C 53.3; N.C.
7.91 / 375 (x464|x468) (x0.946|x0.944)</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td><td>NOT APPLICABLE</td></tr>
<tr><td>14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td><td>N/A / N/A N/A N/A
N/A / N/A N/A N/A</td><td>NOT APPLICABLE</td></tr>
<tr><td>summary: generation</td><td>205 (x1.21) / 1.64 C 89.7; N.C.</td><td>120 (x0.317) / 10.9 C 76.7; N.C.</td><td>234 (x0.385) / 124 C 1.34e+03; N.C.</td></tr>
<tr><td>summary: wall</td><td>78.9 (x20.7) / 1.7e+03 (x0.659)</td><td>201 (x0.736) / 414 (x0.513)</td><td>491 (x0.747) / 7.52e+03 (x0.316)</td></tr>
</table>

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time marked n.c. (not comparable); no generation ratio is formed. Runtime comparisons use native wall time; a separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

5.8 UFO-SM recurrence versus AmpliCol NLC

UFO-SM recurrence versus AmpliCol NLC (block 1 of 2)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	33.3 (x0.316) 0.227 (x1.18|x0.759)	26.3 (x0.39) 0.662 (x1.02|x0.755)	27.6 (x0.485) 3.5 (x1.28|x0.927)
2 $u\bar{d} \to W^+ + (n-1)g$	32 (x0.322) 0.138 (x1.4|x0.898)	27.6 (x0.379) 0.375 (x1.24|x0.897)	25.7 (x0.498) 1.95 (x1.21|x0.859)
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	29.3 (x0.361) 0.467 (x0.691|x0.486)	27.2 (x0.465) 1.09 (x0.634|x0.463)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	29.2 (x0.374) 0.249 (x0.595|x0.312)	28.6 (x0.447) 0.507 (x0.598|x0.348)
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	28.9 (x0.357) 0.938 (x1.01|x0.759)	27.4 (x0.467) 2.84 (x1.06|x0.908)
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	29.4 (x0.372) 1.7 (x0.937|x0.638)	35.7 (x0.455) 16.4 (x0.818|x0.538)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	28.7 (x0.365) 1.16 (x0.61|x0.36)	29.8 (x0.541) 6.19 (x0.976|x0.68)
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	29.5 (x0.354) 7.99 (x0.779|x0.622)	32.4 (x0.499) 121 (x0.494|x0.387)
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	36.4 (x0.314) 5.65 (x1.07|x0.939)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	32.7 (x0.319)	28.6 (x0.369)	30.1 (x0.459)
summary: wall	0.183 (x1.26)	1.69 (x0.818)	17.6 (x0.604)

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

UFO-SM recurrence versus AmpliCol NLC (block 2 of 2)

ID base process	n=4	n=5
1 $d\bar{d} \to Z + (n-1)g$	46 (x0.3) 31.3 (x0.751|x0.502)	52.2 (x0.325) 368 (x0.713|x0.488)
2 $u\bar{d} \to W^+ + (n-1)g$	47.7 (x0.233) 18.4 (x0.685|x0.491)	46.8 (x0.306) 232 (x0.563|x0.39)
3 $d\bar{d} \to e^+e^- + (n-2)g$	43.3 (x0.286) 5.46 (x0.555|x0.474)	56.2 (x0.231) 45 (x0.404|x0.298)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	56.2 (x0.218) 2.47 (x0.402|x0.289)	47.4 (x0.299) 22.4 (x0.4|x0.337)
5 $d\bar{d} \to ZZ + (n-2)g$	59.6 (x0.218) 15.3 (x0.823|x0.673)	55.7 (x0.256) 118 (x0.825|x0.624)
6 $gg \to t\bar{t} + (n-2)g$	82.2 (x0.194) 217 (x0.611|x0.387)	103 (x0.547) 3.52e+03 (x1.63|x1.15)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	50.3 (x0.283) 57.6 (x0.557|x0.419)	93.4 (x0.243) 714 (x0.796|x0.544)
8 $gg \to gg + (n-2)g$	70.2 (x0.424) 2.03e+03 (x0.751|x0.474)	70.4 (x18) 3.67e+04 (x6.93|x4.9)
9 $d\bar{d} \to ZZZ + (n-3)g$	37.1 (x0.321) 15.1 (x0.921|x0.836)	43 (x0.273) 74.9 (x0.743|x0.657)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	118 (x0.113) 3.93 (x0.783|x0.7)	152 (x0.0812) 8.85 (x0.71|x0.633)
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	86 (x0.143) 11.7 (x0.698|x0.619)	150 (x0.085) 65.2 (x0.534|x0.442)
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	124 (x0.0932) 7.33 (x0.433|x0.394)	152 (x0.0833) 15.6 (x0.346|x0.308)
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	N/A N/A N/A N/A	N/A N/A N/A N/A
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE
summary: generation	68.4 (x0.209)	85.2 (x1.44)
summary: wall	201 (x0.733)	3.49e+03 (x6.24)

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

5.9 UFO-SM recurrence versus AmpliCol full-colour

UFO-SM recurrence versus AmpliCol full-colour (block 1 of 2)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	32 (x0.323) 0.271 (x0.993|x0.633)	27.5 (x0.407) 0.743 (x0.905|x0.677)	25.8 (x0.414) 3.9 (x0.897|x0.659)
2 $u\bar{d} \to W^+ + (n-1)g$	32.4 (x0.317) 0.157 (x1.23|x0.788)	25.7 (x0.404) 0.413 (x1.07|x0.769)	27.2 (x0.612) 2.16 (x0.974|x0.698)
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	27.7 (x0.373) 0.522 (x0.63|x0.45)	29.3 (x0.36) 1.21 (x0.531|x0.407)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	27 (x0.379) 0.273 (x0.539|x0.282)	28.5 (x0.488) 0.561 (x0.709|x0.403)
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	27.5 (x0.378) 1.05 (x0.916|x0.693)	27.7 (x0.387) 3.09 (x0.951|x0.783)
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	27.5 (x0.379) 1.97 (x0.818|x0.551)	35 (x0.471) 19.5 (x1.1|x0.619)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	29.2 (x0.388) 1.49 (x0.478|x0.28)	30.5 (x0.347) 7.36 (x0.571|x0.388)
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	28 (x0.374) 9.41 (x0.666|x0.53)	32.5 (x0.531) 145 (x0.533|x0.424)
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	37.3 (x0.309) 6.02 (x0.983|x0.879)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	32.2 (x0.32)	27.5 (x0.385)	30.4 (x0.432)
summary: wall	0.214 (x1.08)	1.98 (x0.702)	21 (x0.627)

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

UFO-SM recurrence versus AmpliCol full-colour (block 2 of 2)

ID base process	n=4	n=5
1 $d\bar{d} \to Z + (n-1)g$	68.5 (x0.203) 36.3 (x0.816|x0.516)	79 (x0.245) 471 (x0.544|x0.385)
2 $u\bar{d} \to W^+ + (n-1)g$	77.3 (x0.151) 21 (x0.709|x0.464)	33.9 (x0.476) 283 (x0.524|x0.358)
3 $d\bar{d} \to e^+e^- + (n-2)g$	72.8 (x0.179) 6.07 (x0.514|x0.409)	77.7 (x0.157) 50.9 (x0.43|x0.29)
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	64 (x0.205) 2.76 (x0.446|x0.316)	40.8 (x0.338) 25.9 (x0.393|x0.311)
5 $d\bar{d} \to ZZ + (n-2)g$	70.6 (x0.198) 16.7 (x0.803|x0.646)	81.7 (x0.181) 133 (x0.78|x0.561)
6 $gg \to t\bar{t} + (n-2)g$	96.7 (x0.186) 284 (x0.479|x0.298)	141 (x0.416) 6.58e+03 (x1.08|x0.693)
7 $d\bar{d} \to t\bar{t} + (n-2)g$	85.5 (x0.169) 73.5 (x0.512|x0.387)	99.1 (x0.223) 1.1e+03 (x0.515|x0.334)
8 $gg \to gg + (n-2)g$	79.7 (x0.403) 3.07e+03 (x0.716|x0.391)	92 (x11.5) 9.73e+04 (x3.72|x2.35)
9 $d\bar{d} \to ZZZ + (n-3)g$	37 (x0.31) 15.9 (x0.895|x0.764)	42.7 (x0.288) 79.6 (x0.698|x0.619)
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	118 (x0.112) 4.09 (x0.85|x0.735)	143 (x0.0855) 9.19 (x0.787|x0.704)
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	94.9 (x0.134) 12.7 (x0.712|x0.686)	149 (x0.088) 71 (x0.458|x0.402)
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	118 (x0.103) 7.59 (x0.434|x0.372)	145 (x0.0868) 16.1 (x0.354|x0.294)
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	N/A N/A N/A N/A	N/A N/A N/A N/A
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE
summary: generation	81.9 (x0.183)	93.8 (x1.13)
summary: wall	296 (x0.694)	8.84e+03 (x3.49)

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Original AmpliCol supports at most three open quark lines; beyond that scope its entry is unsupported and valid candidate generation and runtime values are shown as absolute n.c. quantities. Here n.c. means not comparable. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

5.10 Built-in SM compiled JIT O3 versus recurrence JIT O2 LC

Built-in SM compiled JIT O3 versus recurrence JIT O2 LC (block 1 of 3)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	6.52 / 6.51 N/A N/A 0.245 / 0.134 N/A N/A	6.7 / 6.61 N/A N/A 0.683 / 0.279 N/A N/A	6.75 / 6.61 N/A N/A 1.47 / 0.705 N/A N/A
2 $u\bar{d} \to W^+ + (n-1)g$	6.61 / 6.54 N/A N/A 0.193 / 0.136 N/A N/A	6.61 / 7.23 N/A N/A 0.439 / 0.282 N/A N/A	6.59 / 6.57 N/A N/A 1.02 / 0.666 N/A N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	6.52 / 6.68 N/A N/A 0.309 / 0.226 N/A N/A	6.67 / 6.68 N/A N/A 0.642 / 0.373 N/A N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	6.76 / 7.3 N/A N/A 0.165 / 0.354 N/A N/A	6.55 / 6.54 N/A N/A 0.305 / 0.331 N/A N/A
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	6.65 / 6.6 N/A N/A 0.86 / 0.283 N/A N/A	6.74 / 6.7 N/A N/A 2.6 / 0.535 N/A N/A
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	6.98 / 7.19 N/A N/A 0.866 / 0.394 N/A N/A	6.71 / 6.69 N/A N/A 4.6 / 1.31 N/A N/A
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	6.62 / 7.12 N/A N/A 0.595 / 0.274 N/A N/A	6.68 / 6.54 N/A N/A 3.04 / 0.812 N/A N/A
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	6.54 / 7.15 N/A N/A 1.9 / 0.825 N/A N/A	6.79 / 6.66 N/A N/A 8.08 / 4.97 N/A N/A
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	6.97 / 7.52 N/A N/A 4.75 / 0.705 N/A N/A
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	N/A / N/A	N/A / N/A	N/A / N/A
summary: wall	N/A / N/A	N/A / N/A	N/A / N/A

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

Built-in SM compiled JIT O3 versus recurrence JIT O2 LC (block 2 of 3)

ID base process	n=4	n=5	n=6
1 $d\bar{d} \to Z + (n-1)g$	9.96 / 9.36 N/A N/A 4.52 / 2.5 N/A N/A	7.58 / 9.14 N/A N/A 10.8 / 8.81 N/A N/A	12.1 / 10.3 N/A N/A 35.5 / 73.3 N/A N/A
2 $u\bar{d} \to W^+ + (n-1)g$	7 / 8.2 N/A N/A 3.16 / 2.39 N/A N/A	7.44 / 8.95 N/A N/A 11.1 / 12.4 N/A N/A	11.3 / 11 N/A N/A 44.8 / 79.6 N/A N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	7.8 / 11.2 N/A N/A 1.67 / 0.888 N/A N/A	8.5 / 8.46 N/A N/A 4.07 / 3.15 N/A N/A	13.2 / 12 N/A N/A 16.7 / 18.9 N/A N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	6.99 / 9.02 N/A N/A 0.703 / 0.923 N/A N/A	7.9 / 9.46 N/A N/A 2.6 / 2.74 N/A N/A	9.5 / 10.6 N/A N/A 18 / 12 N/A N/A
5 $d\bar{d} \to ZZ + (n-2)g$	9.06 / 9.1 N/A N/A 6.58 / 1.35 N/A N/A	9.43 / 9.11 N/A N/A 15.7 / 5.03 N/A N/A	12.2 / 9.77 N/A N/A 43.4 / 18.3 N/A N/A
6 $gg \to t\bar{t} + (n-2)g$	8.29 / 9.21 N/A N/A 34.6 / 9.01 N/A N/A	10.9 / 10.4 N/A N/A 167 / 39.7 N/A N/A	72.6 / 20.5 N/A N/A 668 / 291 N/A N/A
7 $d\bar{d} \to t\bar{t} + (n-2)g$	10.1 / 9.4 N/A N/A 15.5 / 4.48 N/A N/A	9.85 / 8.72 N/A N/A 69.8 / 13.3 N/A N/A	24.3 / 13.6 N/A N/A 590 / 90.9 N/A N/A
8 $gg \to gg + (n-2)g$	9.83 / 7.98 N/A N/A 38.6 / 22.9 N/A N/A	15.4 / 13.5 N/A N/A 139 / 209 N/A N/A	184 / 165 N/A N/A 448 / 3.39e+03 N/A N/A
9 $d\bar{d} \to ZZZ + (n-3)g$	8.45 / 7.25 N/A N/A 10.8 / 1.32 N/A N/A	7.53 / 7.67 N/A N/A 25.8 / 2.83 N/A N/A	7.57 / 7.79 N/A N/A 62.4 / 9.25 N/A N/A
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	8.27 / 7.88 N/A N/A 3.23 / 2.25 N/A N/A	7.38 / 9.63 N/A N/A 6.23 / 2.71 N/A N/A	9.04 / 10.5 N/A N/A 11.4 / 4.21 N/A N/A
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	8.67 / 8.81 N/A N/A 3.86 / 3.39 N/A N/A	8.51 / 8.94 N/A N/A 43.2 / 8.2 N/A N/A	21.4 / 11.9 N/A N/A 762 / 63 N/A N/A
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	N/A / N/A N/A N/A N/A / N/A N/A N/A
summary: generation	N/A / N/A	N/A / N/A	N/A / N/A
summary: wall	N/A / N/A	N/A / N/A	N/A / N/A

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

Built-in SM compiled JIT O3 versus recurrence JIT O2 LC (block 3 of 3)

ID base process	n=7	n=8	n=9
1 $d\bar{d} \to Z + (n-1)g$	11.7 / 39.3 (x2.32) (x25.1) 63.5 / 430 (x1.73|NOT EXPOSED) (x1.02|NOT EXPOSED)	83.7 / N/A N/A N/A 165 / N/A N/A N/A	1.13e+03 / N/A N/A N/A 414 / N/A N/A N/A
2 $u\bar{d} \to W^+ + (n-1)g$	12 / 45.9 N/A N/A 48 / 507 N/A N/A	65.1 / N/A N/A N/A 119 / N/A N/A N/A	827 / N/A N/A N/A 295 / N/A N/A N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	8.87 / 10.4 N/A N/A 27.2 / 60.4 N/A N/A	11.2 / N/A N/A N/A 59.6 / N/A N/A N/A	81.7 / N/A N/A N/A 149 / N/A N/A N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	9.75 / 12.3 N/A N/A 15.7 / 58 N/A N/A	10.8 / N/A N/A N/A 78.6 / N/A N/A N/A	60 / N/A N/A N/A 103 / N/A N/A N/A
5 $d\bar{d} \to ZZ + (n-2)g$	10.6 / 23.3 N/A N/A 95.5 / 99.8 N/A N/A	31.1 / N/A N/A N/A 244 / N/A N/A N/A	202 / N/A N/A N/A 506 / N/A N/A N/A
6 $gg \to t\bar{t} + (n-2)g$	2.06e+03 / 408 N/A N/A 6.36e+03 / 7.7e+03 N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
7 $d\bar{d} \to t\bar{t} + (n-2)g$	164 / 70 N/A N/A 1.24e+03 / 1.03e+03 N/A N/A	3.8e+03 / N/A N/A N/A 5.88e+03 / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
8 $gg \to gg + (n-2)g$	4.4e+03 / N/A N/A N/A 1.63e+03 / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
9 $d\bar{d} \to ZZZ + (n-3)g$	8.84 / 12.4 N/A N/A 153 / 31.4 N/A N/A	14.1 / 62.5 N/A N/A 363 / 177 N/A N/A	64.8 / 1.18e+03 N/A N/A 907 / 1.99e+03 N/A N/A
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	10.6 / 9.87 N/A N/A 22.8 / 11.1 N/A N/A	8.14 / N/A N/A N/A 48.5 / N/A N/A N/A	11.4 / N/A N/A N/A 256 / N/A N/A N/A
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	156 / 23.4 N/A N/A 4.06e+03 / 285 N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
summary: generation	11.7 (x2.32) / 39.3 (x25.1)	N/A / N/A	N/A / N/A
summary: wall	63.5 (x1.73) / 430 (x1.02)	N/A / N/A	N/A / N/A

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

5.11 Built-in SM compiled JIT O3 versus recurrence JIT O2 NLC

Built-in SM compiled JIT O3 versus recurrence JIT O2 NLC (block 1 of 2)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	6.5 N/A 0.258 N/A	6.54 N/A 0.65 N/A	6.55 N/A 2.88 N/A
2 $u\bar{d} \to W^+ + (n-1)g$	6.53 N/A 0.192 N/A	6.91 N/A 0.445 N/A	6.55 N/A 1.63 N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	6.87 N/A 0.305 N/A	6.53 N/A 0.616 N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	6.53 N/A 0.148 N/A	6.57 N/A 0.271 N/A
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	6.53 N/A 0.9 N/A	6.5 N/A 2.82 N/A
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	6.99 N/A 1.56 N/A	6.71 N/A 11.5 N/A
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	6.53 N/A 0.701 N/A	6.61 N/A 4.48 N/A
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	6.7 N/A 6.51 N/A	7.25 N/A 54 N/A
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	6.81 N/A 5.89 N/A
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	N/A	N/A	N/A
summary: wall	N/A	N/A	N/A

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

Built-in SM compiled JIT O3 versus recurrence JIT O2 NLC (block 2 of 2)

ID base process	n=4	n=5
1 $d\bar{d} \to Z + (n-1)g$	7.79 N/A 21.8 N/A	13 N/A 260 N/A
2 $u\bar{d} \to W^+ + (n-1)g$	7.06 N/A 12.6 N/A	10.1 N/A 156 N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	7.56 N/A 3 N/A	8.2 N/A 18.5 N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	8.82 N/A 0.934 N/A	8.8 N/A 7.6 N/A
5 $d\bar{d} \to ZZ + (n-2)g$	10.1 N/A 11.6 N/A	10 N/A 115 N/A
6 $gg \to t\bar{t} + (n-2)g$	10.7 N/A 134 N/A	47.7 N/A 5.71e+03 N/A
7 $d\bar{d} \to t\bar{t} + (n-2)g$	8.33 N/A 31.2 N/A	14.2 N/A 587 N/A
8 $gg \to gg + (n-2)g$	22.1 N/A 1.54e+03 N/A	616 N/A 8.73e+04 N/A
9 $d\bar{d} \to ZZZ + (n-3)g$	7.98 N/A 11.7 N/A	7.69 N/A 51.9 N/A
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	8.91 N/A 2.88 N/A	8.16 N/A 6.38 N/A
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	8.48 N/A 7.36 N/A	8.06 N/A 41.3 N/A
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	8.85 N/A 3.06 N/A	7.5 N/A 6.16 N/A
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	N/A N/A N/A N/A	N/A N/A N/A N/A
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE
summary: generation	N/A	N/A
summary: wall	N/A	N/A

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

5.12 Built-in SM compiled JIT O3 versus recurrence JIT O2 full-colour

Built-in SM compiled JIT O3 versus recurrence JIT O2 full-colour (block 1 of 2)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	6.54 N/A 0.265 N/A	6.71 N/A 0.718 N/A	6.67 N/A 2.9 N/A
2 $u\bar{d} \to W^+ + (n-1)g$	7.88 N/A 0.198 N/A	6.56 N/A 0.441 N/A	6.53 N/A 1.91 N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	6.6 N/A 0.311 N/A	6.54 N/A 0.616 N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	6.59 N/A 0.154 N/A	6.67 N/A 0.273 N/A
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	6.65 N/A 1 N/A	6.6 N/A 2.77 N/A
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	6.63 N/A 1.53 N/A	6.84 N/A 13.2 N/A
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	6.81 N/A 0.697 N/A	6.59 N/A 4.44 N/A
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	6.75 N/A 5.96 N/A	7.34 N/A 57 N/A
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	6.88 N/A 5.82 N/A
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	N/A	N/A	N/A
summary: wall	N/A	N/A	N/A

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

Built-in SM compiled JIT O3 versus recurrence JIT O2 full-colour (block 2 of 2)

ID base process	n=4	n=5
1 $d\bar{d} \to Z + (n-1)g$	8.16 N/A 29.4 N/A	14.1 N/A 304 N/A
2 $u\bar{d} \to W^+ + (n-1)g$	8.87 N/A 15.4 N/A	10 N/A 137 N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	8.28 N/A 3.06 N/A	9.42 N/A 23.8 N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	8.9 N/A 1.1 N/A	8.96 N/A 9.47 N/A
5 $d\bar{d} \to ZZ + (n-2)g$	8.83 N/A 12.8 N/A	9.84 N/A 111 N/A
6 $gg \to t\bar{t} + (n-2)g$	11.9 N/A 140 N/A	45.7 N/A 8.04e+03 N/A
7 $d\bar{d} \to t\bar{t} + (n-2)g$	10 N/A 42 N/A	16.3 N/A 617 N/A
8 $gg \to gg + (n-2)g$	24.5 N/A 2.34e+03 N/A	592 N/A 1.13e+05 N/A
9 $d\bar{d} \to ZZZ + (n-3)g$	8.1 N/A 12.2 N/A	8.6 N/A 60 N/A
10 $d\bar{d} \to e^+e^- ZH + (n-4)g$	9.61 N/A 4.04 N/A	8.64 N/A 7.24 N/A
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	8.66 N/A 7.18 N/A	9.78 N/A 37 N/A
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	8.08 N/A 3.23 N/A	9.4 N/A 7.79 N/A
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	N/A N/A N/A N/A	N/A N/A N/A N/A
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE
summary: generation	N/A	N/A
summary: wall	N/A	N/A

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

5.13 Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 LC

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 LC (block 1 of 3)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	6.52 / 6.51 N/A N/A 0.245 / 0.134 N/A N/A	6.7 / 6.61 N/A N/A 0.683 / 0.279 N/A N/A	6.75 / 6.61 N/A N/A 1.47 / 0.705 N/A N/A
2 $u\bar{d} \to W^+ + (n-1)g$	6.61 / 6.54 N/A N/A 0.193 / 0.136 N/A N/A	6.61 / 7.23 N/A N/A 0.439 / 0.282 N/A N/A	6.59 / 6.57 N/A N/A 1.02 / 0.666 N/A N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	6.52 / 6.68 N/A N/A 0.309 / 0.226 N/A N/A	6.67 / 6.68 N/A N/A 0.642 / 0.373 N/A N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	6.76 / 7.3 N/A N/A 0.165 / 0.354 N/A N/A	6.55 / 6.54 N/A N/A 0.305 / 0.331 N/A N/A
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	6.65 / 6.6 N/A N/A 0.86 / 0.283 N/A N/A	6.74 / 6.7 N/A N/A 2.6 / 0.535 N/A N/A
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	6.98 / 7.19 N/A N/A 0.866 / 0.394 N/A N/A	6.71 / 6.69 N/A N/A 4.6 / 1.31 N/A N/A
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	6.62 / 7.12 N/A N/A 0.595 / 0.274 N/A N/A	6.68 / 6.54 N/A N/A 3.04 / 0.812 N/A N/A
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	6.54 / 7.15 N/A N/A 1.9 / 0.825 N/A N/A	6.79 / 6.66 N/A N/A 8.08 / 4.97 N/A N/A
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	6.97 / 7.52 N/A N/A 4.75 / 0.705 N/A N/A
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	N/A / N/A	N/A / N/A	N/A / N/A
summary: wall	N/A / N/A	N/A / N/A	N/A / N/A

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 LC (block 2 of 3)

ID base process	n=4	n=5	n=6
1 $d\bar{d} \to Z + (n-1)g$	9.96 / 9.36 N/A N/A 4.52 / 2.5 N/A N/A	7.58 / 9.14 N/A N/A 10.8 / 8.81 N/A N/A	12.1 / 10.3 N/A N/A 35.5 / 73.3 N/A N/A
2 $u\bar{d} \to W^+ + (n-1)g$	7 / 8.2 N/A N/A 3.16 / 2.39 N/A N/A	7.44 / 8.95 N/A N/A 11.1 / 12.4 N/A N/A	11.3 / 11 N/A N/A 44.8 / 79.6 N/A N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	7.8 / 11.2 N/A N/A 1.67 / 0.888 N/A N/A	8.5 / 8.46 N/A N/A 4.07 / 3.15 N/A N/A	13.2 / 12 N/A N/A 16.7 / 18.9 N/A N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	6.99 / 9.02 N/A N/A 0.703 / 0.923 N/A N/A	7.9 / 9.46 N/A N/A 2.6 / 2.74 N/A N/A	9.5 / 10.6 N/A N/A 18 / 12 N/A N/A
5 $d\bar{d} \to ZZ + (n-2)g$	9.06 / 9.1 N/A N/A 6.58 / 1.35 N/A N/A	9.43 / 9.11 N/A N/A 15.7 / 5.03 N/A N/A	12.2 / 9.77 N/A N/A 43.4 / 18.3 N/A N/A
6 $gg \to t\bar{t} + (n-2)g$	8.29 / 9.21 N/A N/A 34.6 / 9.01 N/A N/A	10.9 / 10.4 N/A N/A 167 / 39.7 N/A N/A	72.6 / 20.5 N/A N/A 668 / 291 N/A N/A
7 $d\bar{d} \to t\bar{t} + (n-2)g$	10.1 / 9.4 N/A N/A 15.5 / 4.48 N/A N/A	9.85 / 8.72 N/A N/A 69.8 / 13.3 N/A N/A	24.3 / 13.6 N/A N/A 590 / 90.9 N/A N/A
8 $gg \to gg + (n-2)g$	9.83 / 7.98 N/A N/A 38.6 / 22.9 N/A N/A	15.4 / 13.5 N/A N/A 139 / 209 N/A N/A	184 / 165 N/A N/A 448 / 3.39e+03 N/A N/A
9 $d\bar{d} \to ZZZ + (n-3)g$	8.45 / 7.25 N/A N/A 10.8 / 1.32 N/A N/A	7.53 / 7.67 N/A N/A 25.8 / 2.83 N/A N/A	7.57 / 7.79 N/A N/A 62.4 / 9.25 N/A N/A
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	8.27 / 7.88 N/A N/A 3.23 / 2.25 N/A N/A	7.38 / 9.63 N/A N/A 6.23 / 2.71 N/A N/A	9.04 / 10.5 N/A N/A 11.4 / 4.21 N/A N/A
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	8.67 / 8.81 N/A N/A 3.86 / 3.39 N/A N/A	8.51 / 8.94 N/A N/A 43.2 / 8.2 N/A N/A	21.4 / 11.9 N/A N/A 762 / 63 N/A N/A
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	N/A / N/A N/A N/A N/A / N/A N/A N/A
summary: generation	N/A / N/A	N/A / N/A	N/A / N/A
summary: wall	N/A / N/A	N/A / N/A	N/A / N/A

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 LC (block 3 of 3)

ID base process	n=7	n=8	n=9
1 $d\bar{d} \to Z + (n-1)g$	11.7 / 39.3 N/A N/A 63.5 / 430 N/A N/A	83.7 / N/A N/A N/A 165 / N/A N/A N/A	1.13e+03 / N/A N/A N/A 414 / N/A N/A N/A
2 $u\bar{d} \to W^+ + (n-1)g$	12 / 45.9 N/A N/A 48 / 507 N/A N/A	65.1 / N/A N/A N/A 119 / N/A N/A N/A	827 / N/A N/A N/A 295 / N/A N/A N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	8.87 / 10.4 N/A N/A 27.2 / 60.4 N/A N/A	11.2 / N/A N/A N/A 59.6 / N/A N/A N/A	81.7 / N/A N/A N/A 149 / N/A N/A N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	9.75 / 12.3 N/A N/A 15.7 / 58 N/A N/A	10.8 / N/A N/A N/A 78.6 / N/A N/A N/A	60 / N/A N/A N/A 103 / N/A N/A N/A
5 $d\bar{d} \to ZZ + (n-2)g$	10.6 / 23.3 N/A N/A 95.5 / 99.8 N/A N/A	31.1 / N/A N/A N/A 244 / N/A N/A N/A	202 / N/A N/A N/A 506 / N/A N/A N/A
6 $gg \to t\bar{t} + (n-2)g$	2.06e+03 / 408 N/A N/A 6.36e+03 / 7.7e+03 N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
7 $d\bar{d} \to t\bar{t} + (n-2)g$	164 / 70 N/A N/A 1.24e+03 / 1.03e+03 N/A N/A	3.8e+03 / N/A N/A N/A 5.88e+03 / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
8 $gg \to gg + (n-2)g$	4.4e+03 / N/A N/A N/A 1.63e+03 / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
9 $d\bar{d} \to ZZZ + (n-3)g$	8.84 / 12.4 N/A N/A 153 / 31.4 N/A N/A	14.1 / 62.5 N/A N/A 363 / 177 N/A N/A	64.8 / 1.18e+03 N/A N/A 907 / 1.99e+03 N/A N/A
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	10.6 / 9.87 N/A N/A 22.8 / 11.1 N/A N/A	8.14 / N/A N/A N/A 48.5 / N/A N/A N/A	11.4 / N/A N/A N/A 256 / N/A N/A N/A
13 $d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	156 / 23.4 N/A N/A 4.06e+03 / 285 N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	N/A / N/A N/A N/A N/A / N/A N/A N/A	N/A / N/A N/A N/A N/A / N/A N/A N/A	NOT APPLICABLE
summary: generation	N/A / N/A	N/A / N/A	N/A / N/A
summary: wall	N/A / N/A	N/A / N/A	N/A / N/A

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

5.14 Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 NLC

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 NLC (block 1 of 2)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	6.5 N/A 0.258 N/A	6.54 N/A 0.65 N/A	6.55 N/A 2.88 N/A
2 $u\bar{d} \to W^+ + (n-1)g$	6.53 N/A 0.192 N/A	6.91 N/A 0.445 N/A	6.55 N/A 1.63 N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	6.87 N/A 0.305 N/A	6.53 N/A 0.616 N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	6.53 N/A 0.148 N/A	6.57 N/A 0.271 N/A
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	6.53 N/A 0.9 N/A	6.5 N/A 2.82 N/A
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	6.99 N/A 1.56 N/A	6.71 N/A 11.5 N/A
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	6.53 N/A 0.701 N/A	6.61 N/A 4.48 N/A
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	6.7 N/A 6.51 N/A	7.25 N/A 54 N/A
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	6.81 N/A 5.89 N/A
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	N/A	N/A	N/A
summary: wall	N/A	N/A	N/A

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 NLC (block 2 of 2)

ID	base process	n=4	n=5
1	$d\bar{d} \to Z + (n-1)g$	7.79 N/A 21.8 N/A	13 N/A 260 N/A
2	$u\bar{d} \to W^+ + (n-1)g$	7.06 N/A 12.6 N/A	10.1 N/A 156 N/A
3	$d\bar{d} \to e^+e^- + (n-2)g$	7.56 N/A 3 N/A	8.2 N/A 18.5 N/A
4	$u\bar{d} \to e^+\nu_e + (n-2)g$	8.82 N/A 0.934 N/A	8.8 N/A 7.6 N/A
5	$d\bar{d} \to ZZ + (n-2)g$	10.1 N/A 11.6 N/A	10 N/A 115 N/A
6	$gg \to t\bar{t} + (n-2)g$	10.7 N/A 134 N/A	47.7 N/A 5.71e+03 N/A
7	$d\bar{d} \to t\bar{t} + (n-2)g$	8.33 N/A 31.2 N/A	14.2 N/A 587 N/A
8	$gg \to gg + (n-2)g$	22.1 N/A 1.54e+03 N/A	616 N/A 8.73e+04 N/A
9	$d\bar{d} \to ZZZ + (n-3)g$	7.98 N/A 11.7 N/A	7.69 N/A 51.9 N/A
10	$d\bar{d} \to e^+e^-ZH + (n-4)g$	8.91 N/A 2.88 N/A	8.16 N/A 6.38 N/A
11	$d\bar{d} \to t\bar{t}ZH + (n-4)g$	8.48 N/A 7.36 N/A	8.06 N/A 41.3 N/A
12	$d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	8.85 N/A 3.06 N/A	7.5 N/A 6.16 N/A
13	$d\bar{d} \to u\bar{u}\,s\bar{s} + (n-4)g$	N/A N/A N/A N/A	N/A N/A N/A N/A
14	$d\bar{d} \to u\bar{u}\,s\bar{s}\,c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE
summary: generation		N/A	N/A
summary: wall		N/A	N/A

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

5.15 Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 full-colour

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 full-colour (block 1 of 2)

ID base process	n=1	n=2	n=3
1 $d\bar{d} \to Z + (n-1)g$	6.54 N/A 0.265 N/A	6.71 N/A 0.718 N/A	6.67 N/A 2.9 N/A
2 $u\bar{d} \to W^+ + (n-1)g$	7.88 N/A 0.198 N/A	6.56 N/A 0.441 N/A	6.53 N/A 1.91 N/A
3 $d\bar{d} \to e^+e^- + (n-2)g$	NOT APPLICABLE	6.6 N/A 0.311 N/A	6.54 N/A 0.616 N/A
4 $u\bar{d} \to e^+\nu_e + (n-2)g$	NOT APPLICABLE	6.59 N/A 0.154 N/A	6.67 N/A 0.273 N/A
5 $d\bar{d} \to ZZ + (n-2)g$	NOT APPLICABLE	6.65 N/A 1 N/A	6.6 N/A 2.77 N/A
6 $gg \to t\bar{t} + (n-2)g$	NOT APPLICABLE	6.63 N/A 1.53 N/A	6.84 N/A 13.2 N/A
7 $d\bar{d} \to t\bar{t} + (n-2)g$	NOT APPLICABLE	6.81 N/A 0.697 N/A	6.59 N/A 4.44 N/A
8 $gg \to gg + (n-2)g$	NOT APPLICABLE	6.75 N/A 5.96 N/A	7.34 N/A 57 N/A
9 $d\bar{d} \to ZZZ + (n-3)g$	NOT APPLICABLE	NOT APPLICABLE	6.88 N/A 5.82 N/A
10 $d\bar{d} \to e^+e^-ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
12 $d\bar{d} \to e^+e^-e^+e^- + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
summary: generation	N/A	N/A	N/A
summary: wall	N/A	N/A	N/A

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 full-colour (block 2 of 2)

ID base process	n=4	n=5
1 $d\bar{d} \to Z + (n-1)g$	8.16 N/A 29.4 N/A	14.1 N/A 304 N/A
2 $u\bar{d} \to W^{+} + (n-1)g$	8.87 N/A 15.4 N/A	10 N/A 137 N/A
3 $d\bar{d} \to e^{+}e^{-} + (n-2)g$	8.28 N/A 3.06 N/A	9.42 N/A 23.8 N/A
4 $u\bar{d} \to e^{+}\nu_{e} + (n-2)g$	8.9 N/A 1.1 N/A	8.96 N/A 9.47 N/A
5 $d\bar{d} \to ZZ + (n-2)g$	8.83 N/A 12.8 N/A	9.84 N/A 111 N/A
6 $gg \to t\bar{t} + (n-2)g$	11.9 N/A 140 N/A	45.7 N/A 8.04e+03 N/A
7 $d\bar{d} \to t\bar{t} + (n-2)g$	10 N/A 42 N/A	16.3 N/A 617 N/A
8 $gg \to gg + (n-2)g$	24.5 N/A 2.34e+03 N/A	592 N/A 1.13e+05 N/A
9 $d\bar{d} \to ZZZ + (n-3)g$	8.1 N/A 12.2 N/A	8.6 N/A 60 N/A
10 $d\bar{d} \to e^{+}e^{-}ZH + (n-4)g$	9.61 N/A 4.04 N/A	8.64 N/A 7.24 N/A
11 $d\bar{d} \to t\bar{t}ZH + (n-4)g$	8.66 N/A 7.18 N/A	9.78 N/A 37 N/A
12 $d\bar{d} \to e^{+}e^{-}e^{+}e^{-} + (n-4)g$	8.08 N/A 3.23 N/A	9.4 N/A 7.79 N/A
13 $d\bar{d} \to u\bar{u}\, s\bar{s} + (n-4)g$	N/A N/A N/A N/A	N/A N/A N/A N/A
14 $d\bar{d} \to u\bar{u}\, s\bar{s}\, c\bar{c} + (n-6)g$	NOT APPLICABLE	NOT APPLICABLE
summary: generation	N/A	N/A
summary: wall	N/A	N/A

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. A separate execution-attribution ratio appears only when both measurements expose one. Not applicable marks a process/multiplicity combination outside the process-family definition. Not exposed means that a successful wall-time measurement has no separately reported execution attribution; for compiled and eager rows, the wall value remains the authenticated warmed total-evaluator boundary. Future compiled and eager entries additionally show an accumulated absolute evaluator-total value marked T; it is not an execution-attribution ratio. Older entries remain marked not exposed; neither label denotes an unfilled measurement.

6 Z-plus-Jets Evaluator Ladders

The dedicated $d\bar{d} \to Z + (n-1)g$ ladders compare the independent reference with compiled JIT O1, compiled JIT O3, ASM O3, C++ O3, eager-DAG JIT O2, and recurrence JIT O2 evaluator configurations. They retain final-state multiplicities $n = 1, \ldots, 9$ for both the built-in model and the UFO-SM result set. Each row states its evaluator backend and optimization level explicitly; all other workload settings follow the benchmark contract above.

The eager and recurrence rows start from the corresponding portable JIT O2 prepared model. Each layout retains complete flow and helicity coverage; the selected-flow workload sums helicities, while the all-flow workload selects one helicity at runtime. Prepared-model creation is excluded from process-generation time. Every displayed Z-ladder timing ratio uses the matching original `AmpliCol` workload as its denominator, while every non-recurrence `pyAmpliCol` result is compared directly with the matching recurrence result at the stricter cross-mode tolerance. Built-in and UFO-SM recurrence results are likewise compared directly.

Runtime comparisons between these backends also include their vectorization strategy. The JIT rows can use SIMD across phase-space points, whereas the generated ASM and C++ rows are scalar. Their ratios therefore include both code-generation quality and vectorization; they are not comparisons of otherwise identical kernels.

Worked $d\bar{d} \to Zg$ example

For $n = 2$, the explicit process is

$$d(p_1)\,\bar{d}(p_2) \to Z(p_3)\,g(p_4).$$

The report labels source particles in process order. The selected leading-colour reference order for the selected-flow timing is therefore $(2, 4, 1, 3)$: the colour line enters through $\bar{d}$, traverses the gluon, and exits through d, while the colour-singlet Z remains in the public ordering only to keep source labels unambiguous. The corresponding coloured word is $(2, 4, 1)$.

`pyAmpliCol` builds a shared-current representation for the selected process. Source nodes carry particle, momentum, helicity, flavour, and colour-flow labels; composite nodes are identified by their complete quantum state. The selected-flow workload uses the traversal $(2, 4, 1)$ and sums helicities; the all-flow workload sums every LC flow for one selected helicity. Both layouts retain complete physical coverage and are validated against the same reference point before timing.

6.1 Built-in SM dedicated $d\bar{d} \to Z+$ gluon performance

Built-in SM $d\bar{d} \to Z+$ gluon ladder (block 1 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [us/pt]	exec [us/pt]	gen [s]	wall [us/pt]	exec [us/pt]
1 `AmpliCol` reference	31.9	0.0565	NOT EXPOSED	0.000361	0.125	NOT EXPOSED
1 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
1 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
1 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
1 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
1 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
1 recurrence JIT O2	6.52 (x0.204)	0.242 (x4.29)	0.166 (x2.94)	C 6.51; N.C.	0.135 (x1.08)	0.0197 (x0.158)
2 `AmpliCol` reference	25.6	0.366	NOT EXPOSED	0.000914	0.288	NOT EXPOSED
2 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
2 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
2 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
2 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
2 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
2 recurrence JIT O2	6.7 (x0.261)	0.614 (x1.68)	0.472 (x1.29)	C 6.61; N.C.	0.288 (x0.999)	0.0933 (x0.324)
3 `AmpliCol` reference	27.4	1.23	NOT EXPOSED	0.00149	0.834	NOT EXPOSED
3 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
3 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
3 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
3 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
3 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
3 recurrence JIT O2	6.75 (x0.246)	1.83 (x1.49)	1.56 (x1.27)	C 6.61; N.C.	0.637 (x0.763)	0.345 (x0.414)

Original AmpliCol is the denominator. Each pyAmpliCol row reports separate topology-replay and all-flow-union generation and runtime measurements. Parenthesized values are candidate/reference ratios. All-flow generation shows the absolute pyAmpliCol value marked n.c.; n.c. means not comparable because its setup boundary differs from the reference. The wall time is the common runtime observable. Not exposed means that a successful wall measurement has no separately reported execution attribution. For compiled and eager rows, wall time still exposes the authenticated warmed total-evaluator boundary. Future compiled and eager entries also show the accumulated absolute evaluator total marked T; T is not an attribution ratio. Older entries remain marked not exposed; it is not a missing measurement.

Built-in SM $d\bar{d} \to Z+$ gluon ladder (block 2 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [us/pt]	exec [us/pt]	gen [s]	wall [us/pt]	exec [us/pt]
4 AmpliCol reference	64.7	4.16	NOT EXPOSED	0.00227	2.93	NOT EXPOSED
4 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
4 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
4 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
4 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
4 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
4 recurrence JIT O2	9.96 (x0.154)	5.57 (x1.34)	5.16 (x1.24)	C 9.36; N.C.	2.6 (x0.887)	2.14 (x0.729)
5 AmpliCol reference	38.8	12.4	NOT EXPOSED	0.0149	15.2	NOT EXPOSED
5 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
5 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
5 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
5 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
5 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
5 recurrence JIT O2	7.58 (x0.195)	10.3 (x0.825)	9.6 (x0.772)	C 9.14; N.C.	8.69 (x0.571)	7.98 (x0.524)
6 AmpliCol reference	151	157	NOT EXPOSED	0.229	97.2	NOT EXPOSED
6 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
6 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
6 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
6 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
6 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
6 recurrence JIT O2	12.1 (x0.0803)	42.9 (x0.274)	38.7 (x0.247)	C 10.3; N.C.	48.1 (x0.495)	47.1 (x0.484)

Original AmpliCol is the denominator. Each pyAmpliCol row reports separate topology-replay and all-flow-union generation and runtime measurements. Parenthesized values are candidate/reference ratios. All-flow generation shows the absolute pyAmpliCol value marked n.c.; n.c. means not comparable because its setup boundary differs from the reference. The wall time is the common runtime observable. Not exposed means that a successful wall measurement has no separately reported execution attribution. For compiled and eager rows, wall time still exposes the authenticated warmed total-evaluator boundary. Future compiled and eager entries also show the accumulated absolute evaluator total marked T; T is not an attribution ratio. Older entries remain marked not exposed; it is not a missing measurement.

Built-in SM $d\bar{d} \to Z+$ gluon ladder (block 3 of 3)

n setup	selected flow, helicity sum gen [s]	wall [us/pt]	exec [us/pt]	all flows, single helicity gen [s]	wall [us/pt]	exec [us/pt]
7 AmpliCol reference	29.2	91.9	NOT EXPOSED	1.45	853	NOT EXPOSED
7 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
7 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
7 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
7 compiled JIT O3	28 (x0.958)	110 (x1.2)	NOT EXPOSED	C 989; N.C.	450 (x0.527)	NOT EXPOSED
7 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
7 recurrence JIT O2	11.6 (x0.397)	63.4 (x0.69)	61.8 (x0.673)	C 39.6; N.C.	430 (x0.504)	434 (x0.509)
8 AmpliCol reference	63.5	234	NOT EXPOSED	N/A	N/A	NOT EXPOSED
8 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
8 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
8 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
8 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
8 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
8 recurrence JIT O2	83.7 (x1.32)	200 (x0.853)	188 (x0.803)	N/A	N/A	N/A
9 AmpliCol reference	62.5	568	NOT EXPOSED	511	2.7e+05	NOT EXPOSED
9 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
9 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
9 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
9 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
9 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
9 recurrence JIT O2	N/A	N/A	N/A	N/A	N/A	N/A

Original AmpliCol is the denominator. Each pyAmpliCol row reports separate topology-replay and all-flow-union generation and runtime measurements. Parenthesized values are candidate/reference ratios. All-flow generation shows the absolute pyAmpliCol value marked n.c.; n.c. means not comparable because its setup boundary differs from the reference. The wall time is the common runtime observable. Not exposed means that a successful wall measurement has no separately reported execution attribution. For compiled and eager rows, wall time still exposes the authenticated warmed total-evaluator boundary. Future compiled and eager entries also show the accumulated absolute evaluator total marked T; T is not an attribution ratio. Older entries remain marked not exposed; it is not a missing measurement.

6.2 UFO-SM dedicated $d\bar{d} \to Z+$ gluon performance

UFO-SM $d\bar{d} \to Z+$ gluon ladder (block 1 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [us/pt]	exec [us/pt]	gen [s]	wall [us/pt]	exec [us/pt]
1 AmpliCol reference	31.9	0.0565	NOT EXPOSED	0.000361	0.125	NOT EXPOSED
1 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
1 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
1 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
1 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
1 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
1 recurrence JIT O2	10.4 (x0.325)	0.249 (x4.4)	0.172 (x3.04)	C 10.2; N.C.	0.14 (x1.12)	0.0239 (x0.192)
2 AmpliCol reference	25.6	0.366	NOT EXPOSED	0.000914	0.288	NOT EXPOSED
2 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
2 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
2 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
2 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
2 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
2 recurrence JIT O2	10.5 (x0.411)	0.633 (x1.73)	0.499 (x1.36)	C 10.4; N.C.	0.295 (x1.02)	0.0948 (x0.329)
3 AmpliCol reference	27.4	1.23	NOT EXPOSED	0.00149	0.834	NOT EXPOSED
3 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
3 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
3 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
3 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
3 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
3 recurrence JIT O2	17.9 (x0.651)	1.76 (x1.44)	1.52 (x1.24)	C 12.6; N.C.	0.705 (x0.845)	0.385 (x0.462)

Original AmpliCol is the denominator. Each pyAmpliCol row reports separate topology-replay and all-flow-union generation and runtime measurements. Parenthesized values are candidate/reference ratios. All-flow generation shows the absolute pyAmpliCol value marked n.c.; n.c. means not comparable because its setup boundary differs from the reference. The wall time is the common runtime observable. Not exposed means that a successful wall measurement has no separately reported execution attribution. For compiled and eager rows, wall time still exposes the authenticated warmed total-evaluator boundary. Future compiled and eager entries also show the accumulated absolute evaluator total marked T; T is not an attribution ratio. Older entries remain marked not exposed; it is not a missing measurement.

UFO-SM $d\bar{d} \to Z+$ gluon ladder (block 2 of 3)

n setup	selected flow, helicity sum gen [s]	wall [us/pt]	exec [us/pt]	all flows, single helicity gen [s]	wall [us/pt]	exec [us/pt]
4 AmpliCol reference	64.7	4.16	NOT EXPOSED	0.00227	2.93	NOT EXPOSED
4 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
4 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
4 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
4 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
4 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
4 recurrence JIT O2	14.1 (x0.218)	5.61 (x1.35)	5.05 (x1.21)	C 14.1; N.C.	2.7 (x0.921)	2.18 (x0.744)
5 AmpliCol reference	38.8	12.4	NOT EXPOSED	0.0149	15.2	NOT EXPOSED
5 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
5 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
5 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
5 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
5 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
5 recurrence JIT O2	13.5 (x0.347)	10.6 (x0.85)	9.83 (x0.791)	C 28.3; N.C.	16.2 (x1.06)	21.7 (x1.43)
6 AmpliCol reference	151	157	NOT EXPOSED	0.229	97.2	NOT EXPOSED
6 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
6 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
6 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
6 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
6 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
6 recurrence JIT O2	12.2 (x0.0809)	25.5 (x0.163)	24.4 (x0.156)	C 13.5; N.C.	53.1 (x0.546)	50.1 (x0.516)

Original AmpliCol is the denominator. Each pyAmpliCol row reports separate topology-replay and all-flow-union generation and runtime measurements. Parenthesized values are candidate/reference ratios. All-flow generation shows the absolute pyAmpliCol value marked n.c.; n.c. means not comparable because its setup boundary differs from the reference. The wall time is the common runtime observable. Not exposed means that a successful wall measurement has no separately reported execution attribution. For compiled and eager rows, wall time still exposes the authenticated warmed total-evaluator boundary. Future compiled and eager entries also show the accumulated absolute evaluator total marked T; T is not an attribution ratio. Older entries remain marked not exposed; it is not a missing measurement.

UFO-SM $d\bar{d} \to Z+$ gluon ladder (block 3 of 3)

n setup	selected flow, helicity sum gen [s]	selected flow, helicity sum wall [us/pt]	selected flow, helicity sum exec [us/pt]	all flows, single helicity gen [s]	all flows, single helicity wall [us/pt]	all flows, single helicity exec [us/pt]
7 AmpliCol reference	29.2	91.9	NOT EXPOSED	1.45	853	NOT EXPOSED
7 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
7 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
7 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
7 compiled JIT O3	29.8 (x1.02)	126 (x1.37)	NOT EXPOSED	C 989; N.C.	482 (x0.565)	NOT EXPOSED
7 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
7 recurrence JIT O2	17.1 (x0.586)	66.1 (x0.719)	64.2 (x0.699)	C 45.9; N.C.	469 (x0.55)	458 (x0.537)
8 AmpliCol reference	63.5	234	NOT EXPOSED	N/A	N/A	NOT EXPOSED
8 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
8 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
8 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
8 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
8 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
8 recurrence JIT O2	87.6 (x1.38)	202 (x0.863)	193 (x0.825)	N/A	N/A	N/A
9 AmpliCol reference	62.5	568	NOT EXPOSED	511	2.7e+05	NOT EXPOSED
9 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
9 compiled ASM O3	N/A	N/A	N/A	N/A	N/A	N/A
9 compiled C++ O3	N/A	N/A	N/A	N/A	N/A	N/A
9 compiled JIT O3	N/A	N/A	N/A	N/A	N/A	N/A
9 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
9 recurrence JIT O2	N/A	N/A	N/A	N/A	N/A	N/A

Original AmpliCol is the denominator. Each pyAmpliCol row reports separate topology-replay and all-flow-union generation and runtime measurements. Parenthesized values are candidate/reference ratios. All-flow generation shows the absolute pyAmpliCol value marked n.c.; n.c. means not comparable because its setup boundary differs from the reference. The wall time is the common runtime observable. Not exposed means that a successful wall measurement has no separately reported execution attribution. For compiled and eager rows, wall time still exposes the authenticated warmed total-evaluator boundary. Future compiled and eager entries also show the accumulated absolute evaluator total marked T; T is not an attribution ratio. Older entries remain marked not exposed; it is not a missing measurement.

Each LC workload reports process-generation time in seconds and native wall time in microseconds per phase-space point, together with a separately measured execution attribution when exposed. Compiled rows use the labelled JIT, ASM, or C++ backend; eager and recurrence rows use portable JIT O2 prepared models. Parenthesized wall-time and selected-flow generation values compare each pyAmpliCol row with the corresponding AmpliCol workload. For compiled and eager rows, wall time is the authenticated warmed total-evaluator boundary; not exposed refers only to a narrower execution attribution. All-flow generation is shown absolutely and marked n.c.; n.c. means not comparable because its setup boundary differs from the reference.

7 Colour-Singlet Model Ladders

Both colour-singlet ladders use compiled JIT O3 evaluation and the full-colour contract. The scalar-contact ladder retains $n = 2, \ldots, 8$ final scalars and tests high-valence contact decomposition, identical-particle normalization, and source parity without Standard-Model-specific kernels. The scalar-gravity ladder retains $n = 2, \ldots, 4$ final gravitons and exercises spin-two wavefunctions, tensor propagators, and momentum-dependent vertices. Because every external particle in these models is a colour singlet, LC, NLC, and full colour reduce to the same unit colour contraction.

For these ladders, the relative-difference row is $\big||\mathcal{M}|^2_{\mathrm{f64}} - |\mathcal{M}|^2_{\mathrm{hp}}\big| / \max(|\mathcal{M}|^2_{\mathrm{hp}}, 10^{-300})$, where the high-precision value uses the same phase-space point. It is a numerical stability check, not a performance ratio.

7.1 Scalar-contact ladder

	scalar_0 scalar_0 > n*scalar_0						
metric	n=2	n=3	n=4	n=5	n=6	n=7	n=8
generation [s]	N/A	N/A	N/A	N/A	N/A	N/A	N/A
wall [μs/pt]	N/A	N/A	N/A	N/A	N/A	N/A	N/A
execution [μs/pt]	N/A	N/A	N/A	N/A	N/A	N/A	N/A
matrix element	N/A	N/A	N/A	N/A	N/A	N/A	N/A
relative diff. vs hp	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Wall time is the common runtime observable. Execution is a separately measured attribution when exposed; future compiled and eager entries show the accumulated absolute evaluator total marked T when narrower attribution is unavailable. NOT EXPOSED denotes a successful wall measurement, and older entries without T remain valid; neither denotes a missing result.

7.2 Scalar-gravity ladder

	scalar_0 scalar_0 > n*graviton		
metric	n=2	n=3	n=4
generation [s]	N/A	N/A	N/A
wall [μs/pt]	N/A	N/A	N/A
execution [μs/pt]	N/A	N/A	N/A
matrix element	N/A	N/A	N/A
relative diff. vs hp	N/A	N/A	N/A

Wall time is the common runtime observable. Execution is a separately measured attribution when exposed; future compiled and eager entries show the accumulated absolute evaluator total marked T when narrower attribution is unavailable. NOT EXPOSED denotes a successful wall measurement, and older entries without T remain valid; neither denotes a missing result.

8 Interpretation

Performance comparisons are meaningful only for a baseline measured with the same observables, parameters, selectors, and hardware profile. The result tables therefore present validated, architecture-local comparisons; they do not combine measurements from different machines. An N/A entry denotes only a structurally inapplicable cell or observable; every applicable cell contains a validated measurement or documented resource-limit status. Numerical agreement, timing uncertainty, and software revision remain part of the published result set even when a compact table shows only the central timing.

9 Worked $d\bar{d} \to Zgg$ Example

This example illustrates how a familiar tree-level process is represented by reusable off-shell currents. It also fixes the momentum, helicity, and colour conventions used in the performance tables.

9.1 External legs and conventions

The physical process is

$$d(p_1)\,\bar{d}(p_2) \longrightarrow Z(p_3)\,g(p_4)\,g(p_5), \qquad p_1 + p_2 = p_3 + p_4 + p_5. \tag{6}$$

The current recursion uses the all-outgoing convention

$$q_1 = -p_1, \quad q_2 = -p_2, \quad q_i = p_i \ (i = 3, 4, 5), \qquad \sum_{i=1}^{5} q_i = 0. \tag{7}$$

External momenta are supplied in the physical convention of Eq. (6) and crossed to the all-outgoing convention before the recursion is evaluated.

label	physical leg	outgoing state	momentum	states	colour role
1	incoming d	outgoing $\bar{d}$	$q_1 = -p_1$	2	$\bar{\mathbf{3}}$, line end
2	incoming $\bar{d}$	outgoing d	$q_2 = -p_2$	2	$\mathbf{3}$, line start
3	outgoing Z	outgoing Z	$q_3 = p_3$	3	singlet
4	outgoing g	outgoing g	$q_4 = p_4$	2	adjoint insertion
5	outgoing g	outgoing g	$q_5 = p_5$	2	adjoint insertion

There are $2 \times 2 \times 3 \times 2 \times 2 = 48$ physical helicity configurations. Twenty-four vanish identically because the massless quark line permits only chirality-compatible initial-state helicities; they remain explicit zero entries in resolved results.

At leading colour the Z is absent from the colour word. Complete coverage contains the two orderings

$$c_0 = (2, 4, 5, 1), \qquad c_1 = (2, 5, 4, 1). \tag{8}$$

Both are retained during generation, so either flow can be selected at evaluation time.

9.2 Reusable current recursion

For a nonempty source subset I, define its momentum by

$$q_I = \sum_{i \in I} q_i. \tag{9}$$

A one-leg current is the appropriate external spinor, polarization vector, or scalar wavefunction. Composite currents obey the binary recursion

$$J^a_{t,\Omega}(I) = {P_t}^a{}_b(q_I) \sum_{\substack{I = A \dot{\cup} B \\ A,B \neq \varnothing}} \sum_{v\,:\,t_A t_B \to t} \kappa_v \, {V_v}^b{}_{cd}(q_A, q_B) J^c_{t_A,\Omega_A}(A) J^d_{t_B,\Omega_B}(B). \tag{10}$$

Here t is the particle state, P_t is its propagator, V_v is an allowed interaction, and Ω collects the quantum numbers required to identify the current. Only combinations consistent with the model, perturbative order, and colour ordering enter the sum.

Representative terms along the quark line are

$$J_d(2,5) = P_d \, V_{dgd}\big[J_d(2), J_g(5)\big], \tag{11}$$

$$J_g(4,5) = P_g \, V_{ggg}\big[J_g(4), J_g(5)\big], \tag{12}$$

$$J_d(2,4,5) = P_d\Big(V_{dgd}\big[J_d(2,5), J_g(4)\big] + V_{dgd}\big[J_d(2,4), J_g(5)\big] + V_{dgd}\big[J_d(2), J_g(4,5)\big]\Big), \tag{13}$$

$$J_d(2,3,4,5) = P_d \sum_{A \dot{\cup} B = \{2,3,4,5\}} V\big[J(A), J(B)\big]. \tag{14}$$

Equation (14) includes every admissible placement of the Z emission and both gluon insertions. Ordered currents such as $J_d(2,5,4)$ and $J_d(2,4,5)$ remain distinct even though their momentum subset is the same.

The final closure joins the endpoint current to source label 1:

$$\mathcal{A}^{(r)}_{h,c} = C^{(r)}_{\dot{a}a} \, J^{\dot{a}}_{\bar{d}}(1) \, J^a_d(2,3,4,5), \qquad \mathcal{A}_{h,c} = \sum_{r \in G(h,c)} \mathcal{A}^{(r)}_{h,c}. \tag{15}$$

The set $G(h,c)$ contains contributions that must be summed coherently. Identical subcurrents are evaluated once and reused wherever their complete quantum-number and ordering labels agree. This sharing is the essential difference between the current DAG and an independent evaluation of every Feynman diagram.

9.3 Resolved observable and normalization

The two leading-colour layouts described in Sec. 11 evaluate the same complete set of helicity and flow components. One is optimized for a selected flow with summed helicities; the other shares work across all flows for a selected helicity. Their resolved components and their summed matrix element are numerically identical.

For the built-in Standard Model, the normalization used in this example is

$$|\mathcal{M}|^2 = \mathcal{N} \sum_{h,c} w_{h,c} \left| \sum_{r \in G(h,c)} \mathcal{A}_{h,c}^{(r)} \right|^2, \qquad \mathcal{N} = \frac{27\,(4\pi\alpha_s)^2 (8\pi\alpha_{\mathrm{EW}})}{36 \times 2}. \tag{16}$$

The denominator averages over the incoming colour and spin states, while the factor $2 = 2!$ accounts for the identical final-state gluons. Coupling powers, colour weights, coherent groups, and symmetry factors are retained separately so that changing numerical model parameters does not alter the process definition.

10 Shared-Current DAG and Numerical Execution

The current recursion becomes a directed acyclic graph (DAG) when physically identical intermediate currents are shared. This section summarizes the identity and ordering needed for that reuse and relates them to the three execution modes.

10.1 Current identity and reuse

Two contributions share a current only when every physics-relevant label agrees. Schematically,

$$\Omega(J) = \big(t, I, H, s, F, Q, C, O, a, \pi_I\big), \tag{17}$$

where the entries identify the particle species, external subset, helicity and spin ancestry, flavour and additive charges, colour state, perturbative order, auxiliary-current type, and colour-sensitive ordering. The subset determines the current momentum $q_I = \sum_{i\in I} q_i$, while the ordered labels prevent distinct colour traversals from being merged.

Exact identity permits one stored current to serve every amplitude contribution that needs it. A second kind of reuse applies when different currents require the same numerical kernel value up to a known exact factor. Neither form of reuse merges distinct physical helicity or colour-flow components.

10.2 Topological ordering

The graph is evaluated in increasing external-subset size. Every input is therefore available before the current that consumes it. For the process in Sec. 9, the ordering is schematically

$$\boxed{\text{sources}} \longrightarrow \boxed{|I| = 2} \longrightarrow \boxed{|I| = 3} \longrightarrow \boxed{|I| = 4} \longrightarrow \boxed{\text{amplitude closures}}. \tag{18}$$

At each stage, all admissible partitions contributing to the same current are summed coherently before propagation. The final closures join the largest retained currents to the remaining external states. This ordering makes the dependency structure explicit without assigning physical meaning to a particular storage layout.

10.3 Numerical realization

Compiled mode combines many relations in a stage into process-specific multi-output evaluators. Eager mode follows the same process graph using reusable prepared-model kernels. Recurrence mode represents the current and closure sequence directly. All three preserve the same coherent sums, colour contraction, helicity treatment, symmetry weights, and normalization.

For resolved leading-colour evaluation, the public result has the form

$$R_{p,h,f} = \big|\mathcal{M}_{h,f}(p)\big|^2, \qquad \text{shape}(R) = (N_{\text{point}}, N_{\text{helicity}}, N_{\text{flow}}). \tag{19}$$

Structural zeros remain explicit, and summing the non-point axes must reproduce the directly evaluated total within the stated numerical tolerance.

10.4 Timing boundary

The primary runtime quantity is the complete native wall time after caller-language array conversion. It includes source and momentum preparation, the mode-specific numerical evaluation, selectors, closures, and final reduction. Packing performed by the caller is outside this boundary.

Some backends additionally expose a narrower execution attribution. Because its precise boundary is mode-specific, it is reported only as a supplementary quantity and only when measured directly. The complete wall time remains the common observable used for comparisons; an unavailable attribution is marked explicitly rather than inferred.

11 Reusable Leading-Colour Execution Layouts

Leading-colour coverage and its numerical layout are separate concepts. Let $\mathcal{H}$ be the retained helicities and $\mathcal{F}$ the retained physical colour flows. Complete leading-colour coverage represents

$$\mathcal{C}_{\mathrm{LC}} = \mathcal{H} \times \mathcal{F} \tag{20}$$

and can expose every resolved amplitude $\mathcal{A}_{hf}$. For selected subsets $S_H \subseteq \mathcal{H}$ and $S_F \subseteq \mathcal{F}$, the observable is

$$\mathcal{W}(S_H, S_F) = \mathcal{N} \sum_{h \in S_H} \sum_{f \in S_F} w_{hf} \left| \sum_{r \in G(h,f)} \mathcal{A}_{hf}^{(r)} \right|^2 . \tag{21}$$

Here $G(h,f)$ is a coherent contribution group, w_{hf} contains exact symmetry and leading-colour weights, and $\mathcal{N}$ is the process normalization. The layout changes how this sum is scheduled, not its components or normalization.

11.1 Two complete-coverage layouts

The report measures two layouts because selected-flow and all-flow workloads offer different opportunities for sharing:

property	**topology replay**	**all-flow union**
Measured workload	one selected flow, summed helicities	all flows, one selected helicity
Retained physics	every flow and helicity	every flow and helicity
Main reuse	equivalent flow topologies related by exact permutations and factors	currents and closures shared directly across all flows
Availability	compiled, eager, and recurrence	compiled, eager, and recurrence

In topology replay, exact relations partition physical flows into equivalence classes. If $\rho(f)$ represents the class containing f, the calculation records a permutation π_f, an exact factor s_f, and the corresponding weight:

$$\mathcal{A}_{hf}(p) = s_f \, \mathcal{A}_{\pi_f(h),\rho(f)}\big(\pi_f(p)\big) . \tag{22}$$

This avoids constructing an independent current graph for every physical ordering and is suited to selected-flow, helicity-summed evaluation.

The all-flow union instead merges the dependency graphs of all resolved components:

$$D_{\mathrm{union}} = \left(\bigcup_{h \in \mathcal{H},\, f \in \mathcal{F}} D_{hf} \right) / \sim, \tag{23}$$

where $\sim$ identifies currents, kernel values, and closures that are algebraically equal up to retained exact factors. For a selected helicity, one graph traversal then reuses intermediate currents across all flows.

Compiled mode specializes the chosen layout to process-wide stage evaluators. Eager mode applies prepared-model kernels to the same dependency structure, and recurrence mode constructs a compact current schedule directly. These execution choices do not change $\mathcal{H}$, $\mathcal{F}$, or Eq. (21).

11.2 Runtime selection

Flow and helicity selections refer to physical components. A selection may be applied to a complete batch or specified independently for each point; omitting an axis sums all retained entries on that axis. Rows with equal selections can be grouped into contiguous numerical batches. This grouping changes neither the generated coverage nor the resolved result.

11.3 Generation and reference boundaries

The two layouts build different graphs and numerical schedules, so their process-generation times are distinct observables:

$$T_{\mathrm{gen}}^{\mathrm{replay}} \quad \text{and} \quad T_{\mathrm{gen}}^{\mathrm{union}}. \tag{24}$$

Cross-mode comparisons use the same layout on both sides. Preparation of reusable model-level kernels occurs before the process-generation timer and is not included in these table values.

The original-`AmpliCol` reference uses two different routes. For the selected-flow workload, its generation value measures creation of the process library. For the all-flow workload, the reported setup value measures preparation of direct all-flow evaluation. The corresponding runtime measurements implement the same physical workloads as `pyAmpliCol`, but the all-flow setup quantities do not define a like-for-like process-generation speedup. The table therefore reports both quantities absolutely and marks their generation comparison as not comparable.

11.4 NLC and full colour

The alternative layouts apply only at leading colour. NLC and full-colour evaluation construct an amplitude basis and contract it with the appropriate colour metric:

$$\mathcal{W}_{\text{contracted}} = \mathcal{N} \sum_{h} \sum_{a,b} \mathcal{A}^{*}_{ha}\, C_{ab}\, \mathcal{A}_{hb}. \tag{25}$$

These treatments do not expose the physical leading-colour flow axis, so there is no topology-replay or all-flow-union distinction.

12 Compiled, Eager, and Recurrence Execution

The three execution modes differ in how they turn the same current recursion into numerical work. Compiled mode specializes groups of current relations to one process. Eager mode applies reusable model kernels according to a process-specific schedule. Recurrence mode constructs a compact schedule of currents, contributions, and closures directly. These choices change the balance between process-generation cost and runtime work, but not the observable being evaluated.

Recurrence JIT O2 is the default execution mode.

mode	process preparation	evaluation	characteristic balance
Compiled	Builds process-specific stage evaluators	Evaluates a small number of process-wide, multi-output stages	More preparation; greater fusion across a process
Eager	Maps the process graph to reusable prepared-model kernels	Applies those kernels according to the process dependencies	Reuses model work across processes
Recurrence	Builds a compact schedule directly from the current recursion	Evaluates scheduled currents, contributions, and closures	Avoids construction of a generic process DAG

12.1 Common physics contract

All three modes retain the same external states, perturbative orders, helicities, colour components, coherent amplitude groups, symmetry factors, and initial-state averages. In a contracted colour basis their common observable has the schematic form

$$\mathcal{M}_p = \mathcal{N} \sum_h \sum_{a,b} A_{ha}(p)^* \, C_{ab} \, A_{hb}(p), \tag{26}$$

with the corresponding resolved form retained when helicity or leading-colour flow components are requested. Each mode must reproduce both the directly evaluated total and the sum of those resolved components.

Current contributions are accumulated coherently before a propagator or other current transformation is applied. Final closures then form the amplitude components entering Eq. (26). This ordering is common to the three modes and prevents execution strategy from changing the normalization.

12.2 Prepared-model boundary

A prepared model contains reusable numerical information derived from the model's particles, parameters, propagators, and interaction tensors. Eager and recurrence JIT O2 measurements reuse this model-level preparation and add only the process-dependent schedule. Compiled JIT O3 measurements instead include construction and compilation of their process-specific evaluators.

Prepared-model creation occurs outside the process-generation timer. The reported eager and recurrence generation times therefore describe adding a process to an already prepared model, not importing and compiling the model itself.

12.3 Batches, selectors, and portability

Evaluation is batched over phase-space points, allowing the JIT backend to use the point dimension for vectorization. Partitioning a large input batch changes neither the process representation nor the numerical result.

Helicity and colour-flow selectors refer to physical resolved components. A complete leading-colour calculation retains every declared helicity and flow, so selecting one flow with summed helicities or one helicity with all flows does not alter the underlying physics coverage. The two layouts used for these workloads are described in Sec. 11.

Portable JIT state is compiled for the receiving processor when loaded. Target-native ASM and C++ payloads remain architecture-specific, and the exact symbolic representation is retained where higher-precision evaluation is supported.

12.4 Interpretation

Compiled mode may benefit most when process-wide fusion outweighs its additional preparation cost. Eager mode can reduce repeated preparation when many processes share a model, while recurrence provides a compact direct representation of the current construction. The tables compare these trade-offs under identical physics, selector, precision, and validation conditions; no mode is assumed to be universally fastest.

13 Supported UFO and Serialized-JSON Models

`pyAmpliCol` accepts a documented subset of the Universal FeynRules Output (UFO) format and an equivalent serialized-JSON representation. Both sources are normalized before process generation and then use the same current recursion, colour treatment, and numerical interfaces as the built-in models. Unsupported structures are rejected explicitly rather than interpreted approximately.

13.1 Sources and normalization

The two input routes are

$$\{\text{trusted UFO module, serialized JSON}\} \longrightarrow \text{normalized model} \longrightarrow \text{numerical kernels.} \tag{27}$$

A UFO module is executable Python and must therefore come from a trusted source. The serialized JSON form is data-only and is preferable for portable exchange and archival. After loading, both forms describe the same particles, parameters, propagators, interactions, coupling orders, Lorentz tensors, and colour tensors.

Model symbols remain in a model-local namespace, so equally named parameters from different models are not identified accidentally. Normalization also makes component orderings and particle conjugation explicit. A model is accepted for ordinary process generation only when the normalized model satisfies the support boundary below; an unrepresentable requested interaction is rejected explicitly.

13.2 Interactions and exact verification

A UFO vertex is a sum over colour and Lorentz structures,

$$\mathcal{V}(1,\ldots,n) = \sum_{a,b} g_{ab}\, C_a(c_1,\ldots,c_n)\, L_b(s_1,p_1;\ldots;s_n,p_n). \tag{28}$$

Each term retains its coupling-order assignment. The supported Lorentz basis includes scalars, momenta, metrics, Dirac identity and gamma matrices, γ_5, chiral projectors, slashed momenta, and sigma tensors. The supported SU(3) representations are the singlet, fundamental, antifundamental, and adjoint.

Colour projection and symmetry reuse are based on exact normalized expressions. Numerical agreement at a sample point is not used to establish an algebraic identity. When an identity needed only for reuse cannot be proved, the corresponding contributions are evaluated separately. When an identity is required to represent an interaction at all, failure to establish it causes the model or process to be rejected.

Leading-colour singlet subtraction, for example, follows from the fundamental Fierz identity

$$(T^a)_i{}^{\bar{\jmath}}(T^a)_k{}^{\bar{l}} = T_R\left(\delta_i{}^{\bar{l}}\delta_k{}^{\bar{\jmath}} - \frac{1}{N_c}\delta_i{}^{\bar{\jmath}}\delta_k{}^{\bar{l}}\right). \tag{29}$$

It is used only when the colour representation, Lorentz interaction, and vector propagator satisfy the corresponding exact conditions. The same principle applies to vectorlike parity, Yang–Mills relations, and current reflection.

13.3 Contact interactions and propagators

Supported contact interactions are rewritten with non-propagating auxiliary currents only when the original tensor can be reconstructed exactly. The physical coupling appears once, and the rewriting introduces no artificial pole. Colour-singlet scalar contacts provide the validation ladder for this construction. Higher-valence singlet interactions are accepted when their normalized tensor admits the same exact reconstruction.

A coloured four-vector gauge contact is accepted only for the two-f structure

$$C^{a_1a_2a_3a_4} = c\, f^{a_ia_jb} f^{ba_ka_r}, \tag{30}$$

up to exact permutations, with one shared adjoint index and a matching Lorentz factorization. Other coloured contact tensors require a separately supported intermediate colour channel.

Standard scalar, Dirac, vector, and spin-two propagators use the mass, width, gauge, and component ordering declared by the normalized model. A custom UFO propagator is accepted only when its numerator and denominator can be lowered exactly; optimizations that assume a standard propagator are then disabled. Massless spin two is validated by the scalar-gravity model. Massive spin-two support remains experimental because it has not yet been tested across a comparably broad set of models.

13.4 Support boundary

In the table, *conditional* means that the stated exact algebraic condition must be established for the particular model expression. It does not imply acceptance of every algebraically equivalent UFO spelling.

area	**class**	**status**	**boundary**
source	UFO directory / JSON	supported	UFO Python must be trusted; JSON is data-only and both use the same normalized model

area	class	status	boundary
spin	scalar, Dirac, vector	supported	standard wavefunctions, propagators, component orderings, and closures
spin	massless spin two	supported	tensor path validated by the scalar-gravity model
spin	massive spin two	experimental	Fierz–Pauli path is available, but broad model validation is incomplete
spin	UFO ghosts	partial	propagating ghosts require an exactly representable custom propagator and are not broadly qualified
spin	Majorana/FNV, spin 3/2	unsupported	no fermion-number-violating-flow or Rarita–Schwinger current prescription
colour	$1, 3, \bar{3}, 8$ of SU(3)	supported	LC, NLC, and full-colour treatments described in this report
colour	$6, \bar{6}$, extra groups	unsupported	no sextet or multiple-group flow and contraction representation
arity 3	$1, \delta, T, f$ colour	supported	exact colour projection and representable Lorentz components
arity 3	d^{abc}, ϵ, K_6, T_6	unsupported	no supported colour-flow and contraction prescription
arity 4	colour singlet	conditional	exact tensor reconstruction by non-propagating auxiliary currents
arity 4	two-f gauge colour	conditional	one shared adjoint index, trivalent topology, and exact Lorentz reconstruction
arity 4	other coloured tensors	unsupported	no supported auxiliary colour channel
arity > 4	colour-singlet contact	conditional	exact no-pole reconstruction with the original coupling inserted once
arity > 4	coloured contact	unsupported	no general irreducible-channel factorization
Lorentz	normalized tensor basis	supported	every tensor function must normalize with explicit component axes
Lorentz	form factors / unknown functions	unsupported	no portable numerical prescription
propagator	standard $0, \frac{1}{2}, 1, 2$	supported	model mass and width, with explicit component ordering and closure
propagator	custom UFO tensor	conditional	exact component lowering; standard-propagator optimizations are disabled
symmetry	vectorlike/Yang–Mills/reflection	conditional	exact normalized identities over the complete interaction sector used by the process

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